HITACHI

Hitachi Vantara

Performance Guide for VSP 5000 Series and VSP E Series

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SVOS

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Exporting Performance Monitor data

You can use Export Tool to export Hitachi Performance Monitor data for your storage system to the spreadsheet application and other applications.

About Export Tool

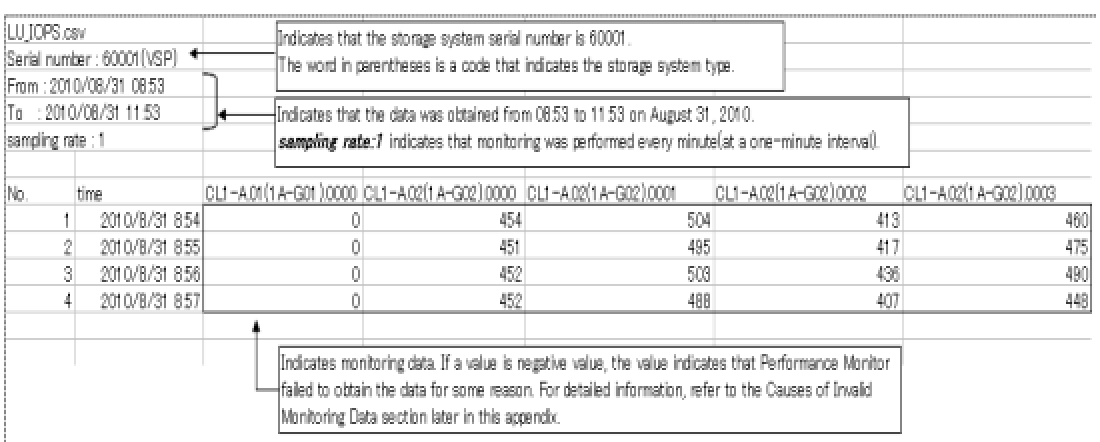
Use Export Tool to export the monitoring data (statistics) shown in the Monitor Performance window to text files. After exporting monitoring data to text files, you can import that data into desktop publishing applications, such as Microsoft Word, or into the spreadsheet application or database applications for analysis. You can also use Export Tool to export monitoring data on remote copy operations performed by the following:

- TrueCopy
- Universal Replicator
- Global-active device
- (VSP 5000 series) TrueCopy for Mainframe
- (VSP 5000 series) Universal Replicator for Mainframe

Example of a text file

The following example is of a text file imported into spreadsheet software.

Translates of Lus (Filename: LU_IOPS.csv)



Note: In this `LU_IOPS.csv` file, the last four digits of a table column heading (such as 0001 and 0002) indicate a LUN. For example, the heading CL1-A.00(1A-G00).0001 indicates the port CL1-A, the host group ID 00, the host group name 1A-G00, and the LUN 0001.

If you export monitoring data about concatenated parity groups, the resulting CSV file does not contain column headings for the concatenated parity groups. For example, if you export monitoring data about a concatenated parity group named 1-3[1-4], you cannot find 1-3[1-4] in column headings. To locate monitoring data about 1-

3[1-4], locate the 1-3 column or the 1-4 column. Either of these columns contains monitoring data about 1-3[1-4].

Preparing to use Export Tool

Export Tool limitation

Running multiple instances of Export Tool simultaneously is not supported. If you run multiple instances, the SVP can become overloaded and a timeout error can occur.

Requirements for Export Tool

The following components are required to use Export Tool (for more information, see the *System Administrator Guide*).

- Windows computer or a UNIX computer

  You can only run Export Tool on Windows computers and UNIX computers that can run Device Manager - Storage Navigator. Both Export Tool and the management client for Device Manager - Storage Navigator use the same supported OS version. For more information about the supported OS version on the management client, see the *System Administrator Guide*.

  Note: If a firewall exists between the Device Manager - Storage Navigator computer and the SVP, see the *System Administrator Guide*. In the section “Setting up TCP/IP for a firewall”, the RMI port numbers listed are the only direct communication settings required for Export Tool.
  You can run Export Tool on the SVP. If the DKCMAIN firmware version is any of the following, the software product media contains the batch file for running Export Tool on the SVP:
  - 93-02-03-x0/00 or later
  - 88-06-02-x0/00 or later
  - 88-05-03-x0/00 or later to 88-05-xx-xx/xx
- Java Runtime Environment (JRE) or OpenJDK

  To use Export Tool, you must have the JRE software or OpenJDK on your Windows computer or UNIX computer. Both Export Tool and Device Manager - Storage Navigator use the same JRE or OpenJDK version. For the required JRE version, see the *System Administrator Guide*.

  If you use the batch file for running Export Tool on the SVP, you do not have to install JRE for Export Tool because necessary JRE has already been installed with the SVP software.
  Note: Export Tool can be installed on the management client.

  (VSP 5000 series) Export Tool uses the Java Runtime Environment (JRE) command and the keytool command. To run these two commands, you must add the path to the commands in your PATH environment variable. For more information about how to add the path to the PATH environment variable, see the documentation for your operating system.

- User ID for exclusive use of Export Tool

  Before you can use Export Tool, you must create a user ID for exclusive use of Export Tool. Assign only the Storage Administrator (Performance Management) role to the user ID for Export Tool. You should not assign any roles other than the Storage Administrator (Performance Management) role or the Storage Administrator (View Only) role that is available by default to this user ID. The user who is assigned to the Storage Administrator (Performance Management) role can perform the following:

  - Save the monitoring data into files
  - Change the sampling interval
  - Start or stop monitoring by using the set subcommand

For details on creating the user ID, see the *System Administrator Guide*.

- Export Tool program

  The media for the software contains Export Tool. For information about how to install Export Tool, see Installing Export Tool. If you want to run Export Tool on the SVP, install Export Tool by following the procedure for installing Export Tool on Windows computers. For the Export Tool version, see the Readme.txt file stored in the software product media.

  Note: Export Tool can be run by using two types of the programs: a program installed from the Host PP disk and a program automatically downloaded from the SVP.
  Each version can be checked by the following:
  - Host PP disk version: specified in ExportTool Ver in Readme.txt.
  - Program version automatically downloaded from the SVP: ExportTool version displayed when running Export Tool.

    This manual refers to the Host PP disk version unless otherwise specified.

Installing Export Tool

Installing Export Tool on Windows

The Export Tool program is a Java class file and is located in the export\lib folder.

CAUTION:
The files edited by the user are overwritten if Export Tool is reinstalled. Save these files in a separate folder in advance.

1. Log on with administrator privileges.
2. Create a new folder for the Export Tool application (for example, C:\Program Files\monitor). If the folder already exists, skip this step.
3. Insert the media into the drive.
4. Locate the \program\monitor\win folder on the media, and copy the self-extracting file export.exe from the media into the new folder you just created.
5. Double-click export.exe to start the installation. However, do not install Export Tool directly under the drive (such as C:\). Export Tool is installed into the installation directory, and a new folder named “export” is created.
   Note: You should delete runUnix.bat, SecurityToolUnix.bat, and delUnix.bat from the export folder because these files are no longer needed.
   Note: If you execute export.exe, Windows might display "This program might not have installed correctly" even though the export tool has been installed correctly. In this case, click "This program installed correctly". If the installation fails, followed by this message, execute export.exe again.

Installing Export Tool on UNIX

The Export Tool program is a Java class file and is located in the lib directory.

CAUTION:
The files edited by the user are overwritten if Export Tool is reinstalled. Save these files in a separate folder in advance.
Note: You should delete the following files from the installation directory because these files are no longer needed:

- runWin.bat, SecurityToolUnix.bat, and delWin.bat
- runWinOnSvp.bat (This file is created only for the following firmware versions):
  - 93-02-03-x0/00 or later
  - 88-06-02-x0/00 or later

- 88-05-03-x0/00 or later to 88-05-xx-xx/xx

1. Log on as a superuser.
   You do not need to remove a previous installation of Export Tool. The new installation overwrites the older program.
2. Create a new directory for the Export Tool program (for example, /monitor).
3. Mount the Export Tool media.
4. Go to the /program/monitor/UNIX directory on the media, and copy the export.tar file to the new directory you just created.
5. Decompress the export.tar file on your computer. Export Tool is installed into the installation directory.

SSL communications using Export Tool

It is recommended to establish the SSL communications when the monitoring data stored on the SVP is transferred to a PC other than the SVP by using Export Tool. The SSL communications make it possible to improve the communications security by encrypting transferred data.

If you have updated a certificate of the SVP, you can verify the validity of the connection destination by registering a root certificate or a self-signed certificate of the SVP in Export Tool. This verification can be enabled or disabled. If the verification cannot be performed when it is enabled, Export Tool will stop the processing. The verification will be performed even when it is disabled, however even if the verification cannot be performed, Export Tool processing will not be suspended.
Note: To use the verification function, make sure to use Export Tool with the DKCMAIN microcode version 90-04-04-00/00 or later and the SVP microcode version 90-04-03/00 or later:

Export Tool operates as a client, while the SVP operates as a server, and therefore, a certificate of the SVP is described as a server certificate. Make sure to use Export Tool with the required version or later, which can be checked in ExportTool Ver. in the Readme.txt file.

If the server certificate on the SVP is a signed public key certificate issued by the Certificate Authority (CA), register the CA root certificate in Export Tool. If the server certificate on the SVP is a self-signed certificate, register the server certificate in Export Tool. The certificate that can be registered in Export Tool must be in X.509 PEM or X.509 DER format. For more information about how to register a root certificate in Export Tool, see Registering a certificate in Export Tool (Windows) and Registering a certificate in Export Tool (UNIX). For more information about how to set a server certificate on the SVP, see the System Administrator Guide.

The following table shows the verification items and the descriptions. The verification is performed for a server certificate of the SVP.

Verification item	Description	Requirement
Expiration date verification	Verify that a certificate has not expired.	Make sure to verify that a server certificate on the SVP has not expired before enabling the verification.
Revocation verification	Verify that a server certificate has not revoked by using the CRL (list of the digital certificates that have been revoked before the expiration date) or OCSP (online verification).	You need a network environment in which the CRL repository or the OCSP responder can be accessed from Export Tool.
SAN or CN verification	Verify that a host name (including FQDN) or an IP address (IPv4 or	The host name or the IP address of the SVP that is specified as a

Verification item	Description	Requirement
	IPv6) entered in subjectAltName (SAN: an additional name of CN) or CommonName (CN) of a server certificate is the same as that of the connected server.	connected server in Export Tool must be included in SAN or CN of the server certificate on the SVP.
Certificate chain verification	Verify a sequence of a root certificate, an intermediate certificate, and a server certificate in a certificate chain.	If a server certificate is signed by an intermediate CA, all of the intermediate certificates including the server certificate must be registered on the SVP.

If a certificate of the SVP has not been updated from the initial state, when the verification function is enabled, the SAN or CN verification cannot be performed, the communication is suspended, and then the Export Tool operation also cannot be performed. Make sure to update the certificate of the SVP before enabling the verification function by following the System Administrator Guide.

For more information about how to enable or disable the verification function, see Java in Export Tool command reference.

Registering a certificate in Export Tool (Windows)

(VSP 5000 series) If a certificate of the SVP has been updated, register a root certificate or a self-signed certificate of the SVP in Export Tool.

1. Log in to the PC running Export Tool.
2. Save the certificate file on the PC running Export Tool.
3. Start the command prompt with administrator permissions.
4. Move the current directory to the directory in which Export Tool is installed.
5. Run the following command: SecurityToolWin.bat import <*SVP-alias*> <*certificate-path*>, where <*SVP-alias*> is any name that indicates the SVP. <*certificate-path*> is the path to the certificate file.
   Example:

```
SecurityToolWin.bat import SVP1 "C:\monitor\cert.crt"
```

Viewing a certificate registered in Export Tool (Windows)

(VSP 5000 series) View a root certificate or a self-signed certificate registered in Export Tool.

1. Log in to the PC running Export Tool.
2. Start the command prompt with administrator permissions.
3. Move the current directory to the directory in which Export Tool is installed.
4. Run the following command: SecurityToolWin.bat list
   Example:

```
Alias name: test
Creation date: Jun 2, 2020
Entry type: trustedCertEntry

Owner: EMAILADDRESS=svp@str.hitachi.co.jp, CN="Hitachi, Ltd.", OU=IT Platfor
m Division Group, O="Hitachi, Ltd.", L=Odawara, ST=Kanagawa, C=JP
```

```
Issuer: EMAILADDRESS=svp@str.hitachi.co.jp, CN="Hitachi, Ltd.", OU=IT Platfo
rm Division Group, O="Hitachi, Ltd.", L=Odawara, ST=Kanagawa, C=JP
Serial number: dc52873fdb5cc76b
Valid from: Thu Apr 17 17:16:04 PDT 2014 until: Wed Apr 17 17:16:04 PDT 2024
Certificate fingerprints:
         MD5:  B3:A5:60:17:17:91:9D:0E:F7:31:DC:1C:06:FA:51:CA
         SHA1: 43:14:DF:80:1D:64:AA:09:B8:F3:1C:13:74:2B:7E:95:1D:2F:E9:6F
         SHA256: 9B:A8:68:45:95:91:3C:72:9B:4C:6A:FE:BB:B9:32:F0:04:E5:9E:D
F:B1:47:2F:59:EA:0C:26:1A:BC:70:E8:15
Signature algorithm name: SHA256withRSA
Version: 1
```

Deleting a certificate from Export Tool (Windows)

(VSP 5000 series) Delete a root certificate or a self-signed certificate registered in Export Tool.

1. Log in to the PC running Export Tool.
2. Verify the SVP alias that is connected by using a root certificate or a self-signed certificate registered in Export Tool. For more information, see [Viewing a certificate registered in Export Tool (Windows)](#).
3. Start the command prompt with administrator permissions.
4. Move the current directory to the directory in which Export Tool is installed.
5. Run the following command: SecurityToolWin.bat delete <*SVP-alias*>, where <*SVP-alias*> is any name that indicates the SVP.
   Example:

   ```
   SecurityToolWin.bat delete SVP1
   ```

Getting help in Export Tool (Windows)

(VSP 5000 series) Get help in Export Tool.

1. Log in to the PC running Export Tool.
2. Start the command prompt with administrator permissions.
3. Move the current directory to the directory in which Export Tool is installed.
4. Run the following command: SecurityToolWin.bat help
   Example:

   ```
   Command Line Syntax
   import <alias> <certificate-file-path>
   alias: alias of specified certificate
   certificate-file-path: relative or absolute certificate file path
   delete <alias>
   alias: alias of specified certificate
   list
   help
   ```

Registering a certificate in Export Tool (UNIX)

(VSP 5000 series) If a certificate of the SVP has been updated, register a root certificate or a self-signed certificate of the SVP in Export Tool.

1. Log in to the PC running Export Tool.
2. Save the certificate file on the PC running Export Tool.
3. Move the current directory to the directory in which Export Tool is installed.

4. Run the following command: SecurityToolUnix.bat import <*SVP-alias*> <*certificate-path*>, where <*SVP-alias*> is any name that indicates the SVP. <*certificate-path*> is the path to the certificate file.
   Example:

   ```
   SecurityToolUnix. bat import SVP1 /tmp/cert.crt
   ```

Viewing a certificate registered in Export Tool (UNIX)

(VSP 5000 series) View a root certificate or a self-signed certificate registered in Export Tool.

1. Log in to the PC running Export Tool.
2. Move the current directory to the directory in which Export Tool is installed.
3. Run the following command: SecurityToolUnix.bat list
   Example:

   ```
   Alias name: test
   Creation date: Jun 3, 2020
   Entry type: trustedCertEntry

   Owner: EMAILADDRESS=svp@str.hitachi.co.jp, CN="Hitachi, Ltd.", OU=IT Platfor
   m Division Group, O="Hitachi, Ltd.", L=Odawara, ST=Kanagawa, C=JP
   Issuer: EMAILADDRESS=svp@str.hitachi.co.jp, CN="Hitachi, Ltd.", OU=IT Platfo
   rm Division Group, O="Hitachi, Ltd.", L=Odawara, ST=Kanagawa, C=JP
   Serial number: dc52873fdb5cc76b
   Valid from: Fri Apr 18 09:16:04 JST 2014 until: Thu Apr 18 09:16:04 JST 2024
   Certificate fingerprints:
            MD5:  B3:A5:60:17:17:91:9D:0E:F7:31:DC:1C:06:FA:51:CA
            SHA1: 43:14:DF:80:1D:64:AA:09:B8:F3:1C:13:74:2B:7E:95:1D:2F:E9:6F
            SHA256: 9B:A8:68:45:95:91:3C:72:9B:4C:6A:FE:BB:B9:32:F0:04:E5:9E:D
   F:B1:47:2F:59:EA:0C:26:1A:BC:70:E8:15
   Signature algorithm name: SHA256withRSA
   Subject Public Key Algorithm: 2048-bit RSA key
   Version: 1
   ```

Deleting a certificate from Export Tool (UNIX)

(VSP 5000 series) Delete a root certificate or a self-signed certificate registered in Export Tool.

1. Log in to the PC running Export Tool.
2. Verify the SVP alias that is connected by using a root certificate or a self-signed certificate registered in Export Tool. For more information, see Viewing a certificate registered in Export Tool (UNIX).
3. Start the command prompt with administrator permissions.
4. Move the current directory to the directory in which Export Tool is installed.
5. Run the following command: SecurityToolUnix.bat delete <*SVP-alias*>, where <*SVP-alias*> is any name that indicates the SVP.
   Example:

   ```
   SecurityToolUnix.bat delete SVP1
   ```

Getting help in Export Tool (UNIX)

(VSP 5000 series) Get help in Export Tool.

1. Log in to the PC running Export Tool.

2. Move the current directory to the directory in which Export Tool is installed.
3. Run the following command: SecurityToolUnix.bat help.
   Example:

```
Command Line Syntax
import <alias> <certificate-file-path>
alias: alias of specified certificate
certificate-file-path: relative or absolute certificate file path
delete <alias>
alias: alias of specified certificate
list
help
```

Using Export Tool

After installing Export Tool, you must prepare a command file and a batch file before you can export any monitoring data.

Preparing a command file

To run Export Tool, you must write scripts for exporting monitoring data. When writing scripts, you must write several subcommands in a command file. When you run Export Tool, the subcommands in the command file run sequentially, and then the monitoring data is saved in files.

When you install Export Tool, the command.txt file is stored in the installation directory. This file contains sample scripts for your command file. You should customize scripts in command.txt according to your needs. For detailed information about subcommand syntax, see Export Tool command reference.

A semicolon (;) indicates the beginning of a comment. Characters from a semicolon to the end of the line are comments.

Example of a command file

(VSP 5000 series)

```
svpip 158.214.135.57          ; Specifies IP address of SVP
login expusr passwd           ; Logs user into the SVP
show                          ; Outputs storing period to standard
                              ; output
group PhyPG Long              ; Specifies type of data to be
                              ; exported and type of
                              ; storing period
group RemoteCopy              ; Specifies type of data to be
                              ; exported
shortrange 201305010850:201305010910
                              ; Specifies term of data to be
                              ; exported for data stored
                              ; in short range
longrange 201304301430:201305011430
                              ; Specifies term of data to be
                              ; exported for data stored
                              ; in long range
outpath out                   ; Specifies directory in which files
                              ; will be saved
option compress               ; Specifies whether to
```

```
                             ; compress files
apply                        ; Executes processing for saving
                             ; monitoring data in files
```

(VSP E series)

```
ip 158.214.135.57:1099         ; Specifies the IP address of
                               ; the SVP and the connection port number
dkcsn 123456                   ; Specifies the target DKC
                               ; serial number
login expusr                   ; Logs the user into the storage system
show                           ; Outputs the storing period in
                               ; the storage system to standard output
group PhyPG                    ; Specifies the type of data to
                               ; be exported and the storing
                               ; period
group RemoteCopy               ; Specifies the type of data to
                               ; be exported
range 200610010850:200610010910  ; Specifies the range in
                               ; which files will be saved
outpath out                    ; Specifies the directory in
                               ; which files will be saved
option compress                ; Specifies whether to
                               ; compress files
apply                          ; Executes processing for
                               ; saving monitoring data in
                               ; files
```

The scripts in this command file are explained as follows:

- (VSP 5000 series) `svpip 158.214.135.57`

  This script specifies that you are logging into the SVP whose IP address is `158.214.135.57`. You must log on to the SVP when using Export Tool.

  The svpip subcommand specifies the IP address of the SVP. You must include the svpip subcommand in your command file. For detailed information about the svpip subcommand, see svpip (VSP 5000 series).

  Export Tool creates a directory with the name specified by the svpip subcommand under the following directory:

  - For Windows: export\lib
  - For UNIX: export\lib

  If the value specified by the svpip subcommand is an IP address, a hexadecimal value is specified as the directory name. Periods (.) and colons (:) cannot be used. If the value is a host name, the IP address of the specified SVP is specified as the directory name.

  The following table provides examples of directory names to be created.

Value specified by the svpip subcommand	Directory name
Value for the svpip subcommand is `svpip 158.214.135.57` (for IPv4)	9ED68739

Value specified by the svpip subcommand	Directory name
Value for the svpip subcommand is `svpip 0000:0000:0020:00B4:0000:0000:9ED6:874` (for IPv6)	00000000002000B4000000009ED68740
Value for the svpip subcommand is `svpip host01` (when the IP address of host01 is 158.214.135.57)	9ED68739

- (VSP E series) `ip 158.214.135.57:1099`

  This script specifies that you are logging into the SVP whose IP address is 158.214.135.57 and that 1099 (the default port of RMIIFRegist) is the port number to connect to the SVP. You must log on to the storage system connected to the SVP when using Export Tool.

  The ip subcommand specifies the SVP to which you wish to connect. You must include the ip subcommand in your command file.

  For detailed information about the ip subcommand, see ip.

  For Export Tool, a directory is created with a value name specified by the ip subcommand under the following directory:

  - For Windows: export\lib
  - For UNIX: export\lib

  When the value specified by the ip subcommand is an IP address, a hexadecimal value is specified as the directory name. Periods (.) and colons (:) are not included. When it is a host name, the IP address of the specified SVP is specified as the directory name. The following table provides examples of the directory names.

Value specified by the ip subcommand	Directory name
Value for the ip subcommand is `ip 158.214.135.57` (for IPv4)	9ED68739
Value for the ip subcommand is `ip 0000:0000:0020:00B4:0000:0000:9ED6:874` (for IPv6)	00000000002000B4000000009ED68740
Value for the ip subcommand is `ip host01` (when the IP address of host01 is 158.214.135.57)	9ED68739

- (VSP E series) `dkcsn 123456`

  This script specifies the serial number of the system from which monitored data will be acquired. This subcommand specifies the system to provide the monitored data. Make sure to notate the dkcsn subcommand in the command file. For detailed information about the dkcsn subcommand, see dkcsn (VSP E series).

- `login expusr passwd`

This script specifies that you provide the user ID expusr and the password passwd to log in to the storage system.

The login subcommand logs the specified user into the storage system. You must include the login subcommand in your command file. For detailed information about the login subcommand, see login.

CAUTION:
When you write the login subcommand in your command file, you must specify a user ID that is used exclusively for running Export Tool. See Requirements for Export Tool for reference.

- `show`

  The show subcommand checks the SVP to find the period of monitoring data stored in the SVP and the data collection interval (called sampling interval in Performance Monitor), and then outputs them to the standard output (for example, the command prompt) and the log file.

  - (VSP 5000 series) Performance Monitor collects statistics by the two types of storing periods: short range and long range. The show subcommand displays the storing periods and the sampling intervals for these two types of monitoring data.

    The following is an example of information that the show subcommand outputs:

    ```
    Short Range     From: 2013/05/01 01:00  -  To: 2013/05/01 15:00
     Interval: 1min.
    Long Range      From: 2013/04/01 00:00  -  To: 2013/05/01 15:00
     Interval: 15min.
    ```

    `Short Range` indicates the storing period and sampling interval of the monitoring data stored in short range. `Long Range` indicates those of the monitoring data stored in long range. In the previous example, the monitoring data in short range is stored every 1 minute in the term of 1:00-15:00 on May 1, 2013. Also, the monitoring data in long range is stored every 15 minutes in the term of April 1, 2013, 0:00 through May 1, 2013, 15:00. When you run Export Tool, you can export monitoring data within these periods into files.

    All of the monitoring items are stored in short range, but a part of monitoring items is stored in both short range and long range. For details on monitoring items that can be stored in long range, see longrange (VSP 5000 series).

  - (VSP E series)

    The following is an example of information that the show subcommand outputs:

    ```
    Range From: 2006/10/01 01:00  -  To: 2006/10/01 15:00
     Interval: 1min.
    ```

    In this example, the monitoring data is stored every 1 minute in the term of 1:00 - 15:00 on October 1, 2006. When you run Export Tool, you can export monitoring data into the file.

  The use of the show subcommand is not mandatory, but you should include the show subcommand in your command file. If an error occurs when you run Export Tool, you might be able to find the error cause by checking the log file for information issued by the show subcommand. For detailed information about the show subcommand, see show.

- `group PhyPG Long` and `group RemoteCopy`

  The group subcommand specifies the type of data that you want to export. Specify an operand following group to define the type of data to be exported. Basically, monitoring data stored in short range is exported. But you can direct the export of monitoring data stored in long range when you specify some of the operands (VSP 5000 series).

(VSP 5000 series) The example script `group PhyPG Long` specifies to export usage statistics about parity groups in long range. Also, the script `group RemoteCopy` specifies to export statistics about remote copy operations by TrueCopy and TrueCopy for Mainframe and monitoring data by global-active device in short range. You can describe multiple lines of the group subcommand to export multiple monitoring items at the same time.

(VSP E series) The example script `group PhyPG` specifies to export usage statistics about parity groups. Also, the script `group RemoteCopy` specifies to export statistics about remote copy operations by TrueCopy and global-active device. You can describe multiple lines of the group subcommand to export multiple monitoring items at the same time.

For detailed information about the group subcommand, see group.

- (VSP 5000 series) `shortrange 201305010850:201310010910` and `longrange 201304301430:201305011430`

  The shortrange subcommand and the longrange subcommand specify the term of monitoring data to be exported. Use these subcommands when you want to narrow the export-target term within the stored data. You can specify both subcommands at the same time. The difference between these subcommands is as follows:

  - The shortrange subcommand is valid for monitoring data in short range. You can use this subcommand to narrow the export-target term for all of the monitoring items you can specify by the group subcommand.

    Specify a term within "`Short Range From` XXX `To` XXX" which is output by the show subcommand.

  - The longrange subcommand is valid for monitoring data in long range. You can use this subcommand only when you specify the PhyPG, PhyLDEV, PhyProc, or PhyMPU operand with the `Long` option in the group subcommand. (The items that can be saved by these operands are the monitoring data displayed on the Physical tab of the Performance Management window with selecting longrange.)

    Specify a term within "`Long Range From` XXX `To` XXX" which is output by the show subcommand.

  In the sample file in Preparing a command file, the script shortrange 201305010850:201305010910 specifies the term 8:50-9:10 on May 1, 2013. This script is applied to the group RemoteCopy subcommand in this example. When you run Export Tool, it will export the statistics about remote copy operations by TrueCopy and TrueCopy for Mainframe and monitoring data by global-active device in the term specified by the shortrange subcommand.

  Also, in Preparing a command file, the script longrange 201304301430:201305011430 specifies the term from April 30, 2013, 14:30 to May 1, 2013, 14:30. This script is applied to the group PhyPG Long subcommand in this example. When you run Export Tool, it will export the usage statistics about parity groups in the term specified by the longrange subcommand.

  If you run Export Tool without specifying the shortrange or longrange subcommand, the monitoring data in the entire storing period (data in the period displayed by the show subcommand) will be exported.

  - For detailed information about the shortrange subcommand, see shortrange (VSP 5000 series).
  - For detailed information about the longrange subcommand, see longrange (VSP 5000 series).

- (VSP E series)`range 200610010850:200610010910`

  The range subcommand specifies the term of monitoring data to be exported. Use these subcommands when you want to narrow the export-target term within the stored data.

You can use this subcommand to narrow the export-target term for all of the monitoring items you can specify by the group subcommand.

Specify a term within `Range From XXX To XXX` which is output by the show subcommand.

In the sample file given above, the script range 200610010850:200610010910 specifies the term 8:50-9:10 on October 1, 2006. This script is applied to the group RemoteCopy subcommand in this example. When you run Export Tool, it will export the statistics about remote copy operations by TrueCopy and global-active device in the term specified by the range subcommand.

If you run Export Tool without specifying the range subcommand, the monitoring data in the whole storing period (data in the period displayed by the show subcommand) will be exported. For detailed information about the range subcommand, see range.

- `outpath out`

  This script specifies that files will be saved in the directory named out in the current directory.

  The outpath subcommand specifies the directory in which files will be saved. For detailed information about the outpath subcommand, see outpath.

- `option compress`

  This script specifies that Export Tool will compress monitoring data in ZIP files.

  The option subcommand specifies whether to save files in ZIP format or in CSV format. For more information, see option.

- `apply`

  The apply subcommand saves monitoring data in files. For detailed information about the apply command, see apply.

Preparing a batch file

Use a batch file to run Export Tool, which starts and saves monitoring data in files when you run the batch file.

The following batch files are stored in the installation directory for Export Tool.

Batch file name	Function	Runs on
runWin.bat	Run Export Tool	Windows computers except the SVP
runWinOnSvp.bat	Run Export Tool (call runWin.bat)	SVP
delWin.bat	Delete directories and files that are created by Export Tool	Windows computers including the SVP
runUnix.bat	Run Export Tool	UNIX computers
delUnix.bat	Delete directories and files that are created by Export Tool	UNIX computers

Note: When you run Export Tool on the SVP, use runWinOnSvp.bat. This batch file calls runWin.bat inside.

The following examples illustrate scripts in runWin.bat and runUnix.bat batch files. These batch files include a command line that runs a Java command. When you run the batch file, the Java command runs the subcommands specified in the command file and then saves monitoring data in files.

Example batch file for Windows computers (runWin.bat):

```
java -classpath "./lib/JSanExportLoader.jar"
-Del.tool.Xmx=536870912 -Dmd.command=command.txt
-Del.logpath=log -Dmd.rmitimeout=20
sanproject.getexptool.RJElMain<CR+LF>
pause<CR+LF>
```

Example batch file for UNIX computers (runUnix.bat):

```
#! /bin/sh<LF>
java -classpath "./lib/JSanExportLoader.jar"
-Del.tool.Xmx=536870912 -Dmd.command=command.txt
-Del.logpath=log -Dmd.rmitimeout=20
sanproject.getexptool.RJElMain<LF>
```

In the previous scripts, <CR+LF> and <LF> indicate the end of a command line.

If you run Export Tool on the SVP for which the SVP software is installed on a directory other than the default directory (C:\Mapp), you need to customize runWinOnSvp.bat. In runWinOnSvp.bat, change the SVP_HOME variable to the installation directory for the SVP software.

```
@echo off
REM This file is a special file for those who want to run ExportTool on SVP.
REM If the installation destination of SVP Software is different from the environ
ment,
REM match the directory of the following SVP_HOME variable with the environment.
setlocal
::::::::::::::::::::
:: Change to your SVP installation directory
set SVP_HOME=C:\Mapp  <-Specify the installation directory of
::::::::::::::::::::     the SVP here.
```

If the computer running Export Tool communicates directly with the SVP, you usually do not need to change scripts in runWin.bat and runUnix.bat. However, you might need to edit the Java command script in a text editor in some occasions, for example:

- If the name of your command file is not command.txt
- If you moved your command file to a different directory
- If you do not want to save in the log directory
- If you want to name log files as you like

If the computer that runs Export Tool communicates with the SVP through a proxy host, edit the Java command script in a text editor to specify the host name (or the IP address) and the port number of the proxy host. For example, if the host name is Jupiter and the port number is 8080, the resulting command script would be as shown in the following examples:

Example of specifying a proxy host on Windows (runWin.bat):

```
java -classpath "./lib/JSanExportLoader.jar"
-Dhttp.proxyHost=Jupiter -Dhttp.proxyPort=8080
-Del.tool.Xmx=536870912 -Dmd.command=command.txt
-Dmd.logpath=log -Dmd.rmitimeout=20 sanproject.getexptool.RJElMain<CR+LF>
pause<CR+LF>
```

Example of specifying a proxy host on UNIX (runUnix.bat):

```
#! /bin/sh<LF>
java -classpath "./lib/JSanExportLoader.jar"
-Dhttp.proxyHost=Jupiter -Dhttp.proxyPort=8080
```

```
-Del.tool.Xmx=536870912
-Dmd.command=command.txt
-Dmd.logpath=log -Dmd.rmitimeout=20 sanproject.getexptool.RJElMain<LF>
```

In the preceding scripts, <CR+LF> and <LF> indicate the end of a command line.

If the IP address of the proxy host is 158.211.122.124 and the port number is 8080, the resulting command script is as follows:

Example batch file for Windows computers (runWin.bat):

```
java -classpath "./lib/JSanExportLoader.jar"
-Dhttp.proxyHost=158.211.122.124
-Dhttp.proxyPort=8080 -Del.tool.Xmx=536870912
-Dmd.command=command.txt
-Dmd.logpath=log -Dmd.rmitimeout=20 sanproject.getexptool.RJElMain<CR+LF>
pause<CR+LF>
```

Example batch file for UNIX computers (runUnix.bat):

```
#! /bin/sh<LF>
java -classpath "./lib/JSanExportLoader.jar"
-Dhttp.proxyHost=158.211.122.124
-Dhttp.proxyPort=8080 -Del.tool.Xmx=536870912
-Dmd.command=command.txt
-Dmd.logpath=log -Dmd.rmitimeout=20 sanproject.getexptool.RJElMain<LF>
```

In the above scripts, <CR+LF> and <LF> indicate the end of a command line.

The following examples illustrate scripts in delWin.bat and delUnix.bat files. These batch files include a command line that runs a Java command. When you run the batch file, the Java command is run, and the directories created by Export Tool, and the files in the directories are deleted.

Example batch file for Windows computers (delWin.bat):

```
java -classpath "./lib/JSanExportLoader.jar"
-Dmd.command=command.txt
-Del.logpath=log
-Del.mode=delete sanproject.getexptool.RJElMain<CR+LF>
```

Example batch file for UNIX computers (delUnix.bat):

```
#! /bin/sh<LF>
java -classpath "./lib/JSanExportLoader.jar"
-Dmd.command=command.txt
-Del.logpath=log
-Del.mode=delete sanproject.getexptool.RJElMain<LF>
```

For detailed information about syntax of the Java command, see Java.

Running Export Tool

Running a batch file

To save monitoring data in files, launch Export Tool by running the batch file.

From a system running Windows, double-click the batch file to run it.

```
c:\WINDOWS> cd c:\export
c:\export> runWin.bat
```

Note: If you run Export Tool on the SVP, run runWinOnSvp.bat.

Dots (...) appear on the screen until the system finishes exporting data. If an internal error occurs, an exclamation mark (!) appears and then Export Tool restarts automatically.

Example of command prompt outputs from Export Tool:

(VSP 5000 series)

```
[ 2] svpip 158.214.135.57
[ 3] login User = expusr, Passwd = [****************]
   :
   :
[ 6] group Port
   :
   :
[20] apply
Start gathering port data
Target = 16, Total = 16
+----+----+----+----+----+----+----+----+----+----+
...........................!
.................................
End gathering port data
```

(VSP E series)

```
Loading ExportTool...
Export tool start [Version 80-xx-xx/xx]
command file = c:\export\command.txt
[ 2] ip 158.214.135.57:1099
[ 3] dkcsn 123456
[ 4] login User = expusr, Passwd = [****************]
   :
   :
[ 6] group Port
   :
   :
[20] apply
Start gathering port data
Target = 16, Total = 16
+----+----+----+----+----+----+----+----+----+----+
...........................!
.................................
End gathering port data
```

It might take time after Loading ExportTool is displayed in the command prompt and before Export tool start [Version 90-*xx-xx*/*xx*] is displayed. The time elapsed between the two messages varies depending on the communication environment of the computer executing Export Tool and SVP. The following table shows the approximate time period.

Communication speed between a computer and SVP	Line usage rate (%)	Approximate time
1 Gbps	0.1	1 minute
	0.5	12 seconds
100 Mbps	0.2	5 minutes
	1	1 minute

By default, the system compresses monitoring data files into a ZIP-format archive file. When you want to view the monitoring data, you can decompress and extract the CSV files from the ZIP archive. If the operating system on your computer does not include a way to extract files from a ZIP archive, you must obtain software to view the data.

When Export Tool is running, if an internal error listed in the Errors in the Export Tool table occurs, an exclamation mark (!) appears, and then Export Tool tries to export again. By default, Export Tool retries three times. If export processing does not end after three retries, or an internal error that is not listed in the table occurs, no retries are performed. In this case, close the command prompt, and then run Export Tool again.

Note that the maximum number of retries can be changed by the retry subcommand. For details about the retry subcommand, see retry.

When Export Tool processing is completed, the directories and files created by Export Tool are deleted automatically. If Export processing stops abnormally, the following directories and files are not deleted:

(VSP 5000 series)

```
export/lib/SVP-value
JSanExport.jar
JSanRmiApiEx.jar
JSanRmiApiSx.jar
JSanRmiServerEx.jar
JSanRmiServerSx.jar
JSanRmiServerUx.jar
```

(VSP E series)

```
export/lib/ip-value
JSanExport.jar
JSanRmiApiSx.jar
JSanRmiServerUx.jar
SanRmiApi.jar
```

To delete the remaining directories and files, run a batch file. If your computer runs Windows, run delWin.bat. If your computer runs UNIX, use delUnix.bat.

CAUTION:
Do not run delWin.bat or delUnix.bat while obtaining monitoring data by running Export Tool.
Note: If you change the ip value (VSP E series) or svpip value (VSP 5000 series) in command.txt before running delWin.bat or delUnix.bat, the directories and files created by Export Tool are not deleted. In this case,

delete them manually in the lib directory.

Note: You can change the default method of exporting files to an uncompressed format. However, the resulting files could be significantly larger and take longer to compile. For more information, see option.

For a complete list of files to be saved by Export Tool, see Using Export Tool.

File formats

If you specify the `nocompress` operand for the option subcommand, Export Tool saves files in CSV format instead of ZIP format (For detailed information, see option.) When files are saved in CSV format instead of ZIP format, the file saving process could take longer and the resulting files could be larger.

Processing time

Files saved by Export Tool are often very large. The total file size can be as large as 2 GB. Therefore, you might need a lot of time to complete the exporting process. If you want to export statistics spanning a long period of time, you should run Export Tool multiple times for shorter periods, rather than run it one time to export the entire time span as a single large file. For example, if you want to export statistics spanning 24 hours, run the tool eight times to export statistics in three-hour increments.

The estimated time that includes the transfer time of the network might take a lot of time, depending on the transmission speed of the network.To shorten the acquisition time, specify an operand of the group subcommand to narrow acquisition objects. For details about the group subcommand, see group. The following table lists time estimates for exporting monitoring data files using different operands in the group subcommand.

Operand for the group subcommand	Estimated time	Description
Port	5 minutes	This estimate assumes that Export Tool is saving 24 hours of statistics for 128 ports.*
PortWWN	5 minutes	This estimate assumes that Export Tool is saving 24 hours of statistics for 128 ports.*
LDEV	60 minutes	This estimate assumes that: • Export Tool is saving 24 hours of statistics for 8,192 volumes.* • Export Tool is used eight times. Each time Export Tool is used, the tool obtains statistics for a 3-hour period.
LU	60 minutes	This estimate assumes that: • Export Tool is saving 24 hours of statistics for 12,288 LUs.* • Export Tool is used eight times. Each time Export Tool is used, the tool obtains statistics for a 3-hour period.
* For a 1-minute interval, the number of hours for the data that is stored is proportional to the interval. For example, for a 2-minute interval, data for 48 hours can be stored.		

Termination code

If you want to use a reference to a termination code in your batch file, perform the following actions:

- To use such a reference in a Windows batch file, write %errorlevel% in the batch file.
- To use such a reference in a UNIX Bourne shell script, write $? in the shell script.
- To use such a reference in a UNIX C shell script, write $status in the shell script.

A reference to a termination code is used in the following example of a Windows batch file. If this batch file runs and Export Tool returns the termination code 1 or 3, the command prompt displays a message that indicates the set subcommand fails.

```
java -classpath "./lib/JSanExportLoader.jar"
-Del.tool.Xmx=536870912 -Dmd.command=command.txt
-Dmd.logpath=log sanproject.getexptool.RJElMain<CR+LF>
if %errorlevel%==1 echo THE SET SUBCOMMAND FAILED<CR+LF>
if %errorlevel%==3 echo THE SET SUBCOMMAND FAILED<CR+LF>
pause<CR+LF>
```

In the previous script, <CR+LF> indicates the end of a command line.

Log files

When Export Tool runs, it creates a log file on your computer. Therefore, if you run Export Tool repeatedly, the size of free space on your computer will be reduced. To secure free space on your computer, it is important that you delete log files regularly. For information about the directory containing log files, see Java.

Export Tool returns a termination code when Export Tool finishes as listed in the following table:

Termination code	Description
0	Export Tool finished successfully.
1	An error occurred when the set subcommand (see set) is run, because an attempt to switch to Modify mode failed. Some other user might have been logged on in Modify mode.
2	Export Tool stops. The following are possible causes: • A command file has been corrupted or could not be read. • An error occurred when a command was parsed. • Maintenance or a configuration change is performed on the SVP. For details about isolating other causes of stops, see Troubleshooting Export Tool for Performance Monitor.
3	An error occurred due to more than one reason. For example, an attempt to switch to Modify mode failed when the set subcommand (see set) is run. Some other user might have been logged on in Modify mode.

Termination code	Description
4	The Storage Administrator (Performance Management) role is not assigned to the user ID.
101	An error occurred during the preparation process for running Export Tool. For details on the error, see the displayed message and the table in Messages issued by Export Tool.

Error handling

When an internal error occurs during export processing, an exclamation mark (!) indicates the error. By default, Export Tool makes up to three more attempts at processing. You can change the maximum number of retries by using the retry subcommand. For detailed information about the retry subcommand, see retry.

If export processing does not finish within three retries or if an internal error occurs other than those listed in the following table, Export Tool stops. If Export Tool stops, quit the command prompt, and then run the tool again.

For more information, see Troubleshooting Export Tool for Performance Monitor.

Errors returned by Export Tool

Error message ID	Description
0001 1001	A timeout error occurred.
0001 4001	An error occurred during the SVP processing.
0001 5400	Because the SVP is busy, the monitoring data cannot be obtained.
0001 5508	An administrator is changing a system environment file.
0002 2016	The array is refreshing, or the settings by the user are registered.
0002 5510	The storage system is in internal process, or some other user is changing configuration.
0002 6502	Now processing.
0002 9000	Another user has locked the file.

Error message ID	Description
0003 2016	A service engineer is accessing the storage system in Modify mode.
0003 2033	The SVP is not ready yet, or an internal processing is being run.
0003 3006	An error occurred during the SVP processing.
0405 6012	An error occurred during the SVP processing.
0405 8003	The storage system status is invalid.
5105 8777	A different user is accessing the storage system in Modify mode.
5205 2003	An internal process is being run, or maintenance is in progress.
5205 2033	The SVP is updating the statistics data.
5305 2033	The SVP is updating the statistics data.
5305 8002	The storage system status is invalid.

Export Tool command reference

Write Export Tool subcommands in your command file and the commands that are used in your batch file. Subcommand list lists and provides links to those subcommands. The Java command is explained in Java.

Export Tool command syntax

The following tables explain the syntax of the Export Tool subcommands that you can write in your command file. The following tables also explain the syntax of the Java command that you should use in your batch file.

- Conventions
- Syntax descriptions
- Writing a script in the command file
- Viewing the online help for subcommands

Conventions

Convention	Description
bold	Indicates characters that you must type exactly as they are shown.
italics	Indicates a type of an operand. You do not need to type characters in italics exactly as they are shown.
[ ]	Indicates one or more operands that you can omit. If two or more operands are enclosed by these square brackets and are delimited by vertical bars (|), you can select one of the operands.
{ }	Indicates that you must select one operand from the operands enclosed by the braces. Two or more operands are enclosed by the braces and are delimited by vertical bars (|).
...	Indicates that a previously used operand can be repeated.
|	Vertical bar delimiter, indicating you can select one of the operands enclosed in square brackets.

Syntax descriptions

This syntax...	Indicates you can write this script...
connect *ip-address*	connect 123.01.22.33
destination [*directory*]	destination destination c:\temp
compress [yes|no]	compress compress yes compress no
answer {yes|no}	answer yes answer no
ports [*name*][...]	ports ports port-1

This syntax...	Indicates you can write this script...
	ports port-1 port-2

Writing a script in the command file

When you write a script in your command file, be aware of the following:

- Ensure that only one subcommand is used in one line.
- Empty lines in any command file will be ignored.
- Use a semicolon (;) if you want to insert a comment in your command file. If you enter a semicolon in one line, the remaining characters in that line will be regarded as a comment.

  Following are examples of comments in a command file:

  (VSP 5000 series)

```
;;;;;;; ;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;
;;;    COMMAND FILE: command.txt    ;;;;
;;;;;;;;;; ;;;;;;;;;;;;;;;;;;;;;;;;;;;;;
svpip 158.214.135.57         ; IP address of the SVP
login expusr "passwd"        ; Log onto the SVP
```

  (VSP E series)

```
;;;;;;; ;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;
;;;    COMMAND FILE: command.txt    ;;;;
;;;;;;;;;; ;;;;;;;;;;;;;;;;;;;;;;;;;;;;;
ip 158.214.135.57:51990      ; IP address of the SVP
dkcsn 123456                 ; Serial No of the DKC
login expusr "passwd"        ; Log into the SVP
```

Viewing the online help for subcommands

You can display the online help to view the syntax of subcommands when you are working at the command prompt. To be able to view the online help, you must use the help subcommand of Export Tool. For more information about how to use the help subcommand, see help.

Subcommand list

Subcommand	Function
svpip (VSP 5000 series) (VSP 5000 series) ip (VSP E series)	Specifies the IP address of the SVP to be logged in.
dkcsn (VSP E series) (VSP E series)	Specifies the serial number of the system which will be used for monitoring data.

Subcommand	Function
retry	Makes settings on retries of export processing.
login	Logs the specified user into the SVP.
show	Checks the SVP to find the period of monitoring data stored in SVP and the data collection interval (sampling interval), and then outputs them to the standard output and the log file.
group	Specifies the type of data that you want to export.
range (VSP E series)	Specifies the term of monitoring data to be exported.
shortrange (VSP 5000 series) (VSP 5000 series)	Specifies the term of monitoring data to be exported for short-range monitoring data.
longrange (VSP 5000 series) (VSP 5000 series)	Specifies the term of monitoring data to be exported for long-range monitoring data.
outpath	Specifies the directory in which files should be saved.
option	Specifies whether to save files in ZIP format or in CSV format.
apply	Saves monitoring data in files.
set	Starts or ends monitoring of the storage system, and specifies the sampling interval.
help	Displays the online help for subcommands.
Java	Starts Export Tool and writes monitoring data into files.

ip

Description

(VSP E series) This subcommand specifies the IP address or the host name of the SVP and the connection port number (the port number of RMIIFRegist). For settings that are affected by changes of port numbers for the SVP, see the *System Administrator Guide*

Syntax

ip {*ip-address*|*host-name*}[:*port-no*]

Operands

Operand	Description
ip-address	Specifies the IP address of SVP. If SVP is managed with IPv6 (Internet Protocol Version 6), you must specify the *IP-address* operand to match the format of IPv6.
host-name	Specifies the host name of SVP. Alphanumeric characters, hyphen, and period can be specified. Underscore (_) cannot be specified. The host name can include a hyphen but must be enclosed by double quotation marks (").
port-no	Specifies the port number of RMIIFRegist for the SVP. Only a number can be specified as a port number. Connect using 1099 when not specifying the port number.

Example

The following example specifies the IP address of the SVP in IPv4 as 158.214.127.170 and the port number as 1099.

```
ip 158.214.127.170:1099
```

The following example specifies the IP address of the SVP in IPv6 as 2001:0DB8:0:CD30:123:4567:89AB:CDEF and the port number as 1099.

```
ip [2001:0DB8:0:CD30:123:4567:89AB:CDEF]:1099
```

svpip (VSP 5000 series)

Description

This subcommand specifies the IP address or the host name of the SVP.

Syntax

svpip {*ip-address*|*host-name*}

Operands

Operand	Description
ip-address	Specifies the IP address of the SVP. If the SVP is managed with IPv6 (Internet Protocol Version 6), you must specify the *IP-address* operand to match the format of IPv6. If Export Tool runs on Windows XP, the interface identifier (for example, "%5") must be added to the end of the specified IP address.
host-name	Specifies the host name of the SVP. Alphanumeric characters, hyphen, and period can be specified. Underscore (_) cannot be specified. The host name can include a hyphen but must be enclosed by double quotation marks (").

Example

The following example specifies the IP address of the SVP as 158.214.127.170.

```
svpip 158.214.127.170
```

dkcsn (VSP E series)

Description

This subcommand specifies the serial number of the system which you use for monitoring data.

Syntax

dkcsn *serial-no*

Operands

Operand	Description
serial-no	Specifies the serial number of the system from which to take monitoring data.

Example

The following example specifies the serial number of the system as 123456.

```
dkcsn 123456
```

retry

Description

This subcommand makes settings on retries of export processing.

When an internal error occurs during export processing, Export Tool stops processing and then retries export processing. By default, Export Tool can retry processing up to three times, but you can change the maximum number of retries by using the retry subcommand.

By default, the interval between one retry and the next retry is two minutes. You can change the interval by using the retry subcommand.

The retry subcommand must run before the login subcommand runs.

Syntax

retry [`time=`*m*] [`count=`*n*]

Operands

Operand	Description
time=*m*	Specifies the interval between retries in minutes, where m is a value within the range of 1 to 59. If this operand is omitted, the interval between retries is two minutes.
count=*n*	Specifies the maximum number of retries. If n is 0, the number of retries is unlimited. If this operand is omitted, the maximum number of retries is 3.

Example

If the following command file is used, the interval between retries is 5 minutes and the maximum number of retries is 10.

(VSP 5000 series)

```
svpip 158.214.135.57
retry time=5 count=10
login expusr passwd
show
group Port
short-range 201304010850:201304010910
outpath
outoption compress
apply
```

(VSP E series)

```
ip 158.214.135.57
dkcsn 123456
retry time=5 count=10
login expusr passwd
show
group Port
range 200604010850:200604010910
```

```
outpath out
option compress
apply
```

login

Description

This subcommand uses a user ID and a password to log the specified user in to the storage system.

(VSP 5000 series) The svpip subcommand must run before the login subcommand runs.

(VSP E series) The ip subcommand must specify the SVP which manages the storage system to log into before the login subcommand runs.

The login subcommand fails if monitoring data does not exist in SVP.

Syntax

login *userid password*

Operands

Operand	Description
userid	(VSP 5000 series) Specifies the user ID for the SVP. (VSP E series) Specifies the exclusive user ID for Export Tool. If the user ID includes any non-alphanumeric character, the user ID must be enclosed by double quotation marks ("). Be sure to specify a user ID that is used exclusively with Export Tool. For detailed information, see Requirements for Export Tool.
password	(VSP 5000 series) Specifies the password to log in to the SVP. (VSP E series) Specifies the exclusive password of the user for Export Tool. The following restrictions apply when you specify the password: • If the password includes any non-alphanumeric character, the password ID must be enclosed by double quotation marks ("). For example, enter "pass&word" in place of pass&word. • If the password includes a back slash (\), add an escape character (\) before the back slash (\), and then enclose the password by double quotation marks ("). For example, enter "pass\\word" in place of pass\word. • If the password includes a double quotation mark ("), add an escape character (\) before the double quotation mark ("), and then enclose the password by double quotation marks ("). For example, enter "pass\"word" in place of pass"word.

Example

This example logs the user expusr into the storage system connected to the SVP whose IP address is 158.214.127.170. The password is pswd.

(VSP 5000 series)

```
svpip 158.214.127.170 login expuser pswd
```

(VSP E series)

```
ip 158.214.127.170 login expuser pswd
```

show

Description

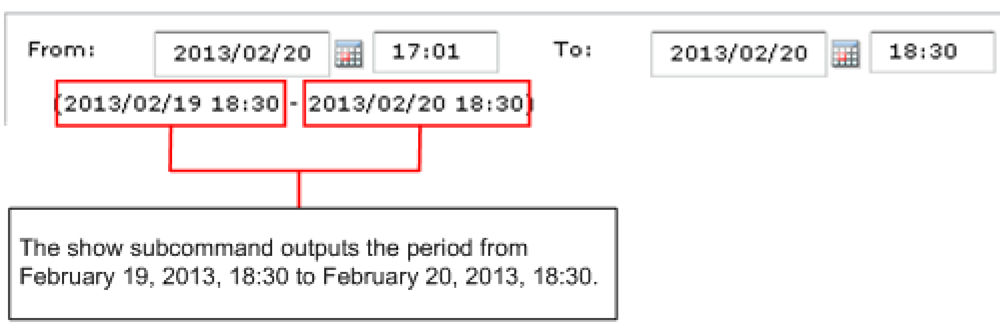
This subcommand outputs the following information to the standard output (for example, to the command prompt):

- The period during which monitoring data was collected onto SVP (storing period).
- The interval at which the monitoring data was collected (sampling interval).

(VSP 5000 series) Performance Monitor collects statistics by the two types of storing periods: in short range and in long range. In short-range monitoring, the monitoring data between 1 day and 15 days is stored in SVP, and in long-range monitoring the monitoring data up to 6 months is stored in SVP. For details about the two storing periods, see shortrange (VSP 5000 series) and longrange (VSP 5000 series).

(VSP E series) Performance Monitor collects data between 1 and 15 days in the SVP.

Storing periods output by the show subcommand are the same as the information displayed in the Monitoring Term area of the Monitor Performance window.



You need to log into the SVP or storage system with the login subcommand before the show subcommand runs.

Syntax

show

Output (VSP 5000 series)

The show subcommand displays the storing period and the sampling interval for these two types of monitoring data: in short range and in long range. For example, the show subcommand outputs the following information:

```
Short Range     From: 2013/05/01 01:00  -  To: 2013/05/01 15:00
 Interval: 1min.
Long Range      From: 2013/04/01 00:00  -  To: 2013/05/01 15:00
 Interval: 15min.
```

`Short Range` indicates the storing period and sampling interval of the monitoring data stored in short range. `Long Range` indicates those of the monitoring data stored in long range. When you run Export Tool, you can export the monitoring data within these periods into files. If you use the shortrange or longrange subcommand additionally, you can narrow the term of data to be exported (see shortrange (VSP 5000 series) or longrange (VSP 5000 series)).

`From` indicates the starting time for collecting monitoring data. To indicates the ending time for collecting monitoring data.

`Interval` indicates the interval at which the monitoring data was collected (sampling interval). For example, `Interval 15min.` indicates that monitoring data was collected at 15-minute intervals.

Output (VSP E series)

The show subcommand displays the storing period and the sampling interval for the monitoring data. For example, the show subcommand outputs the following information:

```
Range     From: 2006/10/01 01:00  -  To: 2006/10/01 15:00
 Interval: 1min.
```

`From` indicates the starting time for collecting monitoring data. To indicates the ending time for collecting monitoring data.

`Interval` indicates the interval at which the monitoring data was collected (sampling interval). For example, `Interval 15min.` indicates that monitoring data was collected at 15-minute intervals.

group

Description

The group subcommand specifies the type of monitoring data that you want to export. This command uses an operand (for example, `PhyPG` and `PhyLDEV` above) to specify a type of monitoring data.

The following table shows the monitoring data that can be saved into files by each operand, and the saved ZIP files. For details on the monitoring data saved in these files, see the tables listed in the See column.

Operand	GUI operation	Monitoring data saved in the file	Saved ZIP file	See
PhyPG	Select Parity Groups from Object list in Performance Objects field in Monitor Performance window.	Usage statistics about parity groups	PhyPG_dat.ZIP [1]	Resource usage and write-pending rate statistics

Operand	GUI operation	Monitoring data saved in the file	Saved ZIP file	See
PhyLDEV	Select Logical Device/Base from Object list in Performance Objects field in Monitor Performance window.	Usage statistics about volumes	PhyLDEV_dat.ZIP [1]	
PhyExG		Usage conditions about external volume groups	PhyExG_dat.ZIP	
PhyExLDEV		Usage conditions about external volumes	PhyExLDEV_dat/PHY_ExLDEV_XXXXX.ZIP [2]	
PhyProc	Select Controller from Object list in Performance Objects field in Monitor Performance window.	Usage statistics about MPs and data recovery and reconstruction processors	PhyProc_dat.ZIP [1]	
PhyProcDetail	Select xx from Object list in Performance Objects field in Monitor Performance window.	Usage statistics about MPs for each resource allocated to MP units	PhyProcDetail_dat.ZIP	MP units
PhyMPU	Select Access Path from Object list in Performance Objects field in Monitor Performance window.	Usage statistics about the following: • (VSP 5000 series) access paths • write pending rate • cache	PhyMPU_dat.ZIP	Resource usage and write-pending rate statistics

Operand	GUI operation	Monitoring data saved in the file	Saved ZIP file	See
PG	Select Parity Group from Object list in Performance Objects field in Monitor Performance window.	Statistics about parity groups, external volume groups	PG_dat.ZIP	Parity group and external volume group statistics
LDEV	Select Logical Device from Object list in Performance Objects field in Monitor Performance window.	Statistics about volumes in parity groups, in external volume groups	LDEV_dat/LDEV_XXXXX.ZIP[3]	Statistics for volumes in parity/external volume groups
LDEVEachOfCU	Select Logical Device/Base from Object list in Performance Objects field in Monitor Performance window.	Statistics about volumes in parity groups or in external volume groups (for volumes controlled by a particular CU)	LDEVEachOfCU_dat/LDEV_XXXXX.ZIP [3]	Volumes in parity groups or external volume groups (at volumes controlled by a particular CU)
Port	Select Fibre Port from Object list in Performance Objects field in Monitor Performance window.	Statistics about ports	Port_dat.ZIP	Port statistics

Operand	GUI operation	Monitoring data saved in the file	Saved ZIP file	See
MFPort (VSP 5000 series)	Select Fibre Port from Object list in Performance Objects field in Monitor Performance window.	Mainframe fibre ports	MF_Port_dat.ZIP	Port statistics
PortWWN	Select WWN from Object list in Performance Objects field in Monitor Performance window.	Statistics about host bus adapters connected to ports	PortWWN_dat.ZIP	Host bus adapters connected to ports statistics
LU	Select LUN/Base from Object list in Performance Objects field in Monitor Performance window.	Statistics about LUs	LU_dat.ZIP	Volumes (LU) statistics
PPCGWWN	Select WWN from Object list in Performance Objects field in Monitor Performance window.	All host bus adapters that are connected to ports	PPCGWWN_dat.ZIP	All host bus adapters connected to ports

Operand	GUI operation	Monitoring data saved in the file	Saved ZIP file	See
RemoteCopy	Select Entire Storage System TC/TCMF/GAD from Object list in Performance Objects field in Monitor Performance window	Statistics about remote copy operations by TrueCopy and TrueCopy for Mainframe and monitoring data by global-active device (in complete volumes)	RemoteCopy_dat.ZIP	Remote copy operations by TC/TCz and monitoring data by GAD (whole volumes)
RCLU	Select LUN TC/GAD from Object list in Performance Objects field in Monitor Performance window	Statistics about remote copy operations by TrueCopy and monitoring data by global-active device (for each volume (LU)). (VSP 5000 series) Note that this data cannot be obtained from a mainfame volume.	RCLU_dat.ZIP	Remote copy operations by TC and monitoring data by GAD (for each volume (LU))
RCLDEV	Select Logical Device TC/TCMF/GAD from Object list in Performance Objects field in Monitor Performance window	Statistics about remote copy operations by TrueCopy and TrueCopy for Mainframe and monitoring data by global-active device (for volumes controlled by a particular CU)	RCLDEV_dat/RCLDEV_XXXXX.ZIP[4]	Remote copy by TC/TCz and monitoring data by GAD (volumes controlled by a particular CU)

Operand	GUI operation	Monitoring data saved in the file	Saved ZIP file	See
Universal Replicator	Select Entire Storage System UR/URMF from Object list in Performance Objects field in Monitor Performance window	Statistics about remote copy operations by Universal Replicator and Universal Replicator for Mainframe (for entire volumes)	UniversalReplicator.ZIP	Remote copy by UR and URz (whole volumes)
URJNL	Select Journal from Object list in Performance Objects field in Monitor Performance window	Statistics about remote copy operations by Universal Replicator and Universal Replicator for Mainframe (for journals)	URJNL_dat.ZIP	Remote copy by UR and URz (at journals)
URLU	Select LUN from Object list in Performance Objects field in Monitor Performance window	Statistics about remote copy operations by Universal Replicator (for each volume (LU)). (VSP 5000 series) Note that this data cannot be obtained from a mainfame volume.	URLU_dat.ZIP	Remote copy by UR (for each volume (LU))

Operand	GUI operation	Monitoring data saved in the file	Saved ZIP file	See
URLDEV	Select Logical Device UR/URMF from Object list in Performance Objects field in Monitor Performance window	Statistics about remote copy operations by Universal Replicator and Universal Replicator for Mainframe (for volumes controlled by a particular CU)	URLDEV_dat/URLDEV_XXXXX.ZIP [5]	Remote copy by UR and URz (at volumes controlled by a particular CU)
PhyMPPK	Select MPPK from Object list in Performance Objects field in Monitor Performance window.	MP usage rate of each resource allocated to MP unit	PhyMPPK_dat.ZIP	MP units

Notes:

1. (VSP 5000 series) When you specify the PhyPG, PhyLDEV, PhyProc, or PhyMPU operand, you can select the storing period of the monitoring data to be exported from short range or long range. When you specify other operands, the monitoring data in short range is exported.
2. A ZIP file name beginning with PhyExLDEV_.
3. A ZIP file name beginning with LDEV_.
4. A ZIP file name beginning with RCLDEV_.
5. A ZIP file name beginning with URLDEV_.

You can use the group subcommand more than one time in a command file. For example, you can write the following script:

```
group PortWWN CL1-A:CL1-B
group RemoteCopy
```

If an operand is used more than one time in a command file, the last operand takes effect. In the example below, the first group subcommand does not take effect, but the second group subcommand takes effect:

```
group PortWWN CL1-A:CL1-B
group PortWWN CL2-A:CL2-B
```

Syntax

(VSP 5000 series)

```
group {PhyPG [Short|Long] [parity-group-id]:[parity-group-id]][ …]|
       PhyLDEV [Short|Long] [parity-group-id]:[parity-group-id]][ …]|
```

```
        PhyExG [[exg-id]:[exg-id]][ …]|
        PhyExLDEV [exg-id]:[exg-id]][ …]|
        PhyProc [Short|Long]|
     PhyProcDetail |
     PhyMPU [Short|Long]|
        PG [[parity-group-id|exg-id:
        [parity-group-id|exg-id]][ …]|
        LDEV [[[parity-group-id|exg-id]:
        [parity-group-id|exg-id]][.…]|
     LDEVEachOfCU[[[LDKC-CU-id]:[LDKC-CU-
id]][ …]|internal|virtual]|
internal|virtual]|
     Port [[port-name]:[port-name]][...]|
     MFPort [[port-name]:[port-name]][...]|
     PortWWN [port-name]:[port-name]][...]|
     LU[[port-name.host-group-id]:[port-name.host-group-id]][ …]|
     PPCGWWN[[monitor-target-name:monitor-target-name]][ …]|
     RemoteCopy|
     RCLU [[port-name.host-group-id]:[port-name.host-group-id]][ …]|
     RCLDEV[[LDKC-CU-id]:[LDKC-CU-id]][ …]|
     UniversalReplicator |
     URJNL[[JNL-group-id]:[JNL-group-id]][ …]|
     URLU [[port-name.host-group-id]:[port-name.host-group-id]][ …]|
     URLDEV[[LDKC-CU-id]:[LDKC-CU-id]][ …]|
        PhyMPPK
        }
```

(VSP E series)

```
group {PhyPG[parity-group-id]:[parity-group-id]][ …]|
        PhyLDEV  [parity-group-id]:[parity-group-id]][ …]|
        PhyExG [[exg-id]:[exg-id]][ …]|
        PhyExLDEV [exg-id]:[exg-id]][ …]|
        PhyProc |
             PG [[parity-group-id|exg-id]:
        [parity-group-id|exg-id]][ …]|
        LDEV [[[parity-group-id|exg-id]:
        [parity-group-id|exg-id]][ …]|
     LDEVEachOfCU[[[LDKC-CU-id]:[LDKC-CU-id]][ …]|internal|virtual]|
internal|virtual]|
     Port [[port-name]:[port-name]][ …]|
     PortWWN [port-name]:[port-name]][ …]|
     LU[[port-name.host-group-id]:
     [port-name.host-group-id]][ …]|
     PPCGWWN[[monitor-target-name:monitor-target-name]][ …]|
     RemoteCopy|
     RCLU [[port-name.host-group-id]:
     [port-name.host-group-id]][ …]|
     RCLDEV[[LDKC-CU-id]:[LDKC-CU-id]][ …]|
     UniversalReplicator |
     URJNL[[JNL-group-id]:[JNL-group-id]][ …]|
     URLU [[port-name.host-group-id]:
     [port-name.host-group-id]][ …]|
     URLDEV [[LDKC-CU-id]:[LDKC-CU-id]][ …]|
        }
```

Operands

Operand	Description
PhyPG [Short|Long] [[*parity-group-id*]:[*parity-group-id*]][…] (VSP 5000 series) PhyPG [[*parity-group-id*]:[*parity-group-id*]][…] (VSP E series)	Use this operand to export statistics about parity group usage rates, which are displayed in the Monitor Performance window. When statistics are exported to a ZIP file, the file name will be PhyPG_dat.ZIP. For details on the statistics exported by this operand, see Resource usage and write-pending rate statistics. (VSP 5000 series) You can use the `Short` or `Long` option to select the storing period of the monitoring data to be exported. If you specify `Short`, the exported file will contain statistics in a short range for up to 15 days. If you specify `Long`, the exported file will contain statistics in a long range for up to six months (for example, up to 186 days). If neither `Short` nor `Long` is specified, statistics in both the short and long range are exported. (VSP E series) The exported file will contain statistics for up to 15 days. When you specify variable *parity-group-id*, you can narrow the range of parity groups whose monitoring data are to be exported. *parity-group-id* is a parity group ID. The colon (:) indicates a range. For example, `1-1:1-5` indicates parity groups from 1-1 to 1-5. Ensure that the *parity-group-id* value on the left of the colon is smaller than the *parity-group-id* value on the right of the colon. For example, you can specify `PhyPG 1-1:1-5`, but you cannot specify `PhyPG 1-5:1-1`. Also, you can specify `PhyPG 1-5:2-1`, but you cannot specify `PhyPG 2-1:1-5`. If the parity group ID is specified only on the right of the colon, all the IDs up to the ID specified on the right will be selected. If the parity group ID is specified only on the left of the colon, the ID specified on the left and all the IDs following it will be selected. If the same ID is specified on both the left and right of the colon, only the specified parity group ID is selected. If *parity-group-id* is not specified, the monitoring data of all parity groups will be exported.
PhyLDEV [Short|Long][[*parity-group-id*]:[*parity-group-id*]][…] (VSP 5000 series) PhyLDEV [[*parity-group-id*]:[*parity-group-id*]][…] (VSP E series)	Use this operand when you want to export statistics about volume usage rates, which are displayed in the Monitor Performance window. When statistics are exported to a ZIP file, the file name will be PhyLDEV_dat.ZIP. For details on the statistics exported by this operand, see Resource usage and write-pending rate statistics. (VSP 5000 series) You can use the `Short` or `Long` option to select the storing period of the monitoring data to be exported. If you specify `Short`, the exported file will contain statistics in short range for up to 15 days. If you specify `Long`, the exported file will contain statistics in long range for up to six months (for example, up to 186 days). If neither `Short` nor `Long` is specified, statistics in both the short and long range are exported. (VSP E series) The exported file will contain statistics for up to 15 days. When you specify variable *parity-group-id*, you can narrow the range of parity groups whose monitoring data are to be exported. *parity-group-id* is a parity group ID. The

Operand	Description
	colon (:) indicates a range. For example, `1-1:1-5` indicates parity groups from 1-1 to 1-5. Ensure that the *parity-group-id* value on the left of the colon is smaller than the *parity-group-id* value on the right of the colon. For example, you can specify `PhyLDEV 1-1:1-5`, but you cannot specify `PhyLDEV 1-5:1-1`. Also, you can specify `PhyLDEV 1-5:2-1`, but you cannot specify `PhyLDEV 2-1:1-5`. If the parity group ID is specified only on the right of the colon, all the IDs up to the ID specified on the right will be selected. If the parity group ID is specified only on the left of the colon, the ID specified on the left and all the IDs following it will be selected. If the same ID is specified on both the left and right of the colon, only the specified parity group ID is selected. If *parity-group-id* is not specified, the monitoring data of all volumes will be exported.
PhyExG [[*exg-id*]:[*exg-id*]][…]	Use this operand when you want to export statistics about external volume groups, which are displayed in the Monitor Performance window. When statistics are exported to a ZIP file, the file name will be PhyExG_dat.ZIP. For details on the statistics exported by this operand, see Resource usage and write-pending rate statistics. When you specify variable *exg-id*, you can narrow the range of external volume groups whose monitoring data are to be exported. *exg-id* is an ID of an external volume group. The colon (:) indicates a range. For example, `E1-1:E1-5` indicates external volume groups from E1-1 to E1-5. Ensure that the *exg-id* value on the left of the colon is smaller than the *exg-id* value on the right of the colon. For example, you can specify `PhyExG E1-1:E1-5`, but you cannot specify `PhyExG E1-5:E1-1`. Also, you can specify `PhyExG E1-5:E2-1`, but you cannot specify `PhyExG E2-1:E1-5`. If *exg-id* is not specified, the monitoring data of all external volume groups will be exported.
PhyExLDEV [[*exg-id*]:[*exg-id*]][…]	Use this operand when you want to export statistics about volumes in external volume groups, which are displayed in the Monitor Performance window. When statistics are exported to a ZIP file, the file name will be PhyExLDEV_dat.ZIP. For details on the statistics exported by this operand, see Resource usage and write-pending rate statistics. When you specify variable *exg-id*, you can narrow the range of external volume groups whose monitoring data are to be exported. *exg-id* is an ID of an external volume group. The colon (:) indicates a range. For example, `E1-1:E1-5` indicates external volume groups from E1-1 to E1-5. Ensure that the *exg-id* value on the left of the colon is smaller than the *exg-id* value on the right of the colon. For example, you can specify `PhyExLDEV E1-1:E1-5`, but you cannot specify `PhyExLDEV E1-5:E1-1`. Also, you can specify `PhyExLDEV E1-5:E2-1`, but you cannot specify `PhyExLDEV E2-1:E1-5`. If *exg-id* is not specified, the monitoring data of all external volumes will be exported.

Operand	Description
PhyProc [Short|Long] (VSP 5000 series) PhyProc (VSP E series)	Use this operand when you want to export the following statistics, which are displayed in the Monitor Performance window: • Usage rates of MPs • (VSP 5000 series) Usage rates of DRRs (data recovery and reconstruction processors) When statistics are exported to a ZIP file, the file name will be PhyProc_dat.ZIP. For details on the statistics exported by this operand, see Resource usage and write-pending rate statistics. (VSP 5000 series) You can use the `Short` or `Long` option to select the storing period of the monitoring data to be exported. If you specify `Short`, the exported file will contain statistics in short range for up to 15 days. If you specify `Long`, the exported file will contain statistics in long range for up to six months (for example, up to 186 days). If neither `Short` nor `Long` is specified, statistics in both the short and long range are exported. (VSP E series) The exported file will contain statistics for up to 15 days.
PhyProcDetail	Use this operand when you want to export the monitoring data of performance information on each MP unit. When statistics are exported to a ZIP file, the file name will be PhyProcDetail_dat.ZIP.
PhyMPU [Short|Long] (VSP 5000 series)	Use this operand when you want to export the following statistics, which are displayed in the Monitor Performance window: • Usage rates of access paths between HIEs and ISWs • Usage rates of access paths between MP units and HIEs • Usage rates of cache memories • Size of the allocated cache memories • Write pending rates When statistics are exported to a ZIP file, the file name will be PhyMPU_dat.ZIP. For details on the statistics exported by this operand, see Resource usage and write-pending rate statistics. You can use the `Short` or `Long` option to select the storing period of the monitoring data to be exported. If you specify `Short`, the exported file will contain statistics in short range for up to 15 days. If you specify `Long`, the exported file will contain statistics in long range for up to six months (for example, up to 186 days). If neither `Short` nor `Long` is specified, statistics in both the short and long range are exported.
PhyMPU (VSP E series)	Use this operand when you want to save the following information displayed in the Monitor Performance window to a file: • Cache memory usage statistics • Amount of allocated cache memory • Write pending rate

Operand	Description
	When statistics are exported to a ZIP file, the file name will be PhyMPU_dat.zip. For details on the statistics exported by this operand, see Resource usage and write-pending rate statistics. The exported file will contain statistics for up to 15 days.
PG [[*parity-group-id*|*exg-id*|*Migration-Volume-group-id*]:[*parity-group-id*|*exg-id*|*Migration-Volume-group-id*]][…] (VSP 5000 series)	Use this operand when you want to export statistics about parity groups, external volume groups, or migration volume groups which are displayed in the Monitor Performance window. When statistics are exported to a ZIP file, the file name will be PG_dat.ZIP. For details on the statistics exported by this operand, see Resource usage and write-pending rate statistics. When you specify variables *parity-group-id*, *exg-id*, or *Migration-Volume-group-id* you can narrow the range of parity groups, external volume groups, or migration volume groups whose monitoring data are to be exported. *parity-group-id* is a parity group ID. *exg-id* is an ID of an external volume group. *Migration-Volume-group-id* is a migration volume group ID. You can check to which group each LDEV belongs in the Logical Devices window. The colon (:) indicates a range. For example, `1-1:1-5` indicates parity groups from 1-1 to 1-5. `E1-1:E1-5` indicates external volume groups from E1-1 to E1-5. M1-1:M5-1 indicates migration volume groups from M1-1 to M5-1. Ensure that the *parity-group-id*, *exg-id*, or *Migration-Volume-group-id* value on the left of the colon is smaller than the *parity-group-id*, *exg-id*, or *Migration-Volume-group-id* value on the right of the colon. For example, you can specify `PG 1-1:1-5`, but you cannot specify `PG 1-5:1-1`. Also, you can specify `PG 1-5:2-1`, but you cannot specify `PG 2-1:1-5`. If the parity group ID is specified only on the right of the colon, all the IDs up to the ID specified on the right will be selected. If the parity group ID is specified only on the left of the colon, the ID specified on the left and all the IDs following it will be selected. If the same ID is specified on both the left and right of the colon, only the specified parity group ID is selected. If none of *parity-group-id*, *exg-id*, or *Migration-Volume-group-id* are specified, the statistics of all parity groups, external volume groups, and migration volume groups will be exported.
PG [[*parity-group-id*|*exg-id*]: [*parity-group-id*|*exg-id*]][…] (VSP E series)	Use this operand when you want to export statistics about parity groups or external volume groups which are displayed in the Monitor Performance window. When statistics are exported to a ZIP file, the file name will be PG_dat.ZIP. For details on the statistics exported by this operand, see Resource usage and write-pending rate statistics. When you specify variables *parity-group-id*, or *exg-id* you can narrow the range of parity groups, or external volume groups whose monitoring data are to be exported. *parity-group-id* is a parity group ID. *exg-id* is an ID of an external volume group. You can check to which group each LDEV belongs in the Logical Devices window. The colon (:) indicates a range. For example, `1-1:1-5` indicates parity groups from 1-1 to 1-5. `E1-1:E1-5` indicates external volume groups from E1-1 to E1-5. Ensure that the *parity-group-id*, or *exg-id* value on the left of the colon is smaller than the *parity-group-id*, or *exg-id* value on the right of the colon. For example, you can

Operand	Description
	specify PG 1-1:1-5, but you cannot specify PG 1-5:1-1. Also, you can specify PG 1-5:2-1, but you cannot specify PG 2-1:1-5. If the parity group ID is specified only on the right of the colon, all the IDs up to the ID specified on the right will be selected. If the parity group ID is specified only on the left of the colon, the ID specified on the left and all the IDs following it will be selected. If the same ID is specified on both the left and right of the colon, only the specified parity group ID is selected. If neither *parity-group-id*, nor *exg-id* is specified, the statistics of all parity groups and external volume groups will be exported.
LDEV [[[*parity-group-id*|*exg-id*|*Migration-Volume-group-id*]:[*parity-group-id*|*exg-id*|*Migration-Volume-group-id*]][…] |internal|virtual]|(VSP 5000 series)	Use this operand when you want to export statistics about volumes, which are displayed in the Monitor Performance window. When statistics are exported to a ZIP file, multiple ZIP files whose names are beginning with LDEV_ will be output. For details on the statistics exported by this operand, see [Port statistics](). When you specify variables *parity-group-id*, *exg-id*, or *Migration-Volume-group-id* you can narrow the range of parity groups, external volume groups, or migration volume groups whose monitoring data are to be exported. *parity-group-id* is a parity group ID. *exg-id* is an ID of an external volume group. *Migration-volume-group-id* is a migration volume group ID. You can check to which group each LDEV belongs in the Logical Devices window. The colon (:) indicates a range. For example, 1-1:1-5 indicates parity groups from 1-1 to 1-5. E1-1:E1-5 indicates external volume groups from E1-1 to E1-5. M1-1:M5-1 indicates migration volume groups from M1-1 to M5-1. Ensure that the *parity-group-id*, *exg-id*, or *Migration-Volume-group-id* value on the left of the colon is smaller than the *parity-group-id*, *exg-id*, or *Migration-Volume-group-id* value on the right of the colon. For example, you can specify LDEV 1-1:1-5, but you cannot specify LDEV 1-5:1-1. Also, you can specify LDEV 1-5:2-1, but you cannot specify LDEV 2-1:1-5. If the parity group ID is specified only on the right of the colon, all the IDs up to the ID specified on the right will be selected. If the parity group ID is specified only on the left of the colon, the ID specified on the left and all the IDs following it will be selected. If the same ID is specified on both the left and right of the colon, only the specified parity group ID is selected. If internal is specified, you can save statistics about volumes in the parity group. If virtual is specified, you can save statistics about volumes in the external volume group or migration volume group. If none of *parity-group-id*, *exg-id*, or *Migration-Volume-group-id* are specified, the statistics of all parity groups, external volume groups, and migration volume groups will be saved to a file. If none of *parity-group-id*, *exg-id*, *Migration-Volume-group-id*, *internal*, or *virtual* is specified as the LDEV operand, data that can be saved when internal or virtual is specified is saved. One of the following values can be specified: • *parity-group-id*, *exg-id*, or *Migration-Volume-group-id* • internal

Operand	Description
	• virtual
LDEV [[[*parity-group-id*|*exg-id*]:[*parity-group-id*||*exg-id*]][…]|internal|virtual]| (VSP E series)	Use this operand when you want to export statistics about volumes, which are displayed in the Monitor Performance window. When statistics are exported to a ZIP file, multiple ZIP files whose names are beginning with LDEV_ will be output. For details on the statistics exported by this operand, see Port statistics. When you specify variables *parity-group-id*, or *exg-id* , you can narrow the range of parity groups, or external volume groups whose monitoring data are to be exported. *parity-group-id* is a parity group ID. *exg-id* is an ID of an external volume group. You can check to which group each LDEV belongs in the Logical Devices window. The colon (:) indicates a range. For example, `1–1:1–5` indicates parity groups from 1-1 to 1-5. `E1–1:E1–5` indicates external volume groups from E1-1 to E1-5. Ensure that the *parity-group-id*, or *exg-id* value on the left of the colon is smaller than the *parity-group-id*, or *exg-id* value on the right of the colon. For example, you can specify `LDEV 1–1:1–5`, but you cannot specify `LDEV 1–5:1–1`. Also, you can specify `LDEV 1–5:2–1`, but you cannot specify `LDEV 2–1:1–5`. If the *parity-group-id* is specified only on the right of the colon, all the IDs up to the ID specified on the right will be selected. If the *parity-group-id* is specified only on the left of the colon, the ID specified on the left and all the IDs following it will be selected. If the same ID is specified on both the left and right of the colon, only the specified *parity-group-id* is selected. If internal is specified, you can save statistics about volumes in the parity group. If virtual is specified, you can save statistics about volumes in the external volume group. If neither *parity-group-id*, nor *exg-id* is specified, the statistics of all parity groups and external volumes will be saved to a file. If none of *parity-group-id*, *exg-id*, *internal*, or *virtual* is specified as the LDEV operand, data that can be saved when internal or virtual is specified is saved. Either one of the following values can be specified: • *parity-group-id*, or *exg-id* • internal • virtual
LDEVEachOfCU [[[*LDKC-CU-id*]:[*LDKC-CU-id*]][…]|internal|virtual]	Use this operand when you want to export statistics about volumes which are displayed in the Monitoring Performance window. By using this operand, you can export monitoring data about volumes controlled by a particular CU. When statistics are exported to a ZIP file, multiple ZIP files whose names are beginning with `LDEV_` will be output. For details on the statistics exported by this operand, see Volumes in parity groups or external volume groups (at volumes controlled by a particular CU). When you specify variable *LDKC-CU-id*, you can narrow the range of LDKC:CUs that control the volumes whose monitoring data are to be exported. *LDKC-CU-id* is an ID of a LDKC:CU. The colon (:) indicates a range. For example, `000:105` indicates LDKC:CUs from 00:00 to 01:05.

Operand	Description
	Ensure that the *LDKC-CU-id* value on the left of the colon is smaller than the *LDKC-CU-id* value on the right of the colon. For example, you can specify `LDEVEachOfCU 000:105`, but you cannot specify `LDEVEachOfCU 105:000`. If internal is specified, you can export statistics about volumes in the parity group. If virtual is specified, you can export statistics about volumes in the external volume or V-VOL. If none of *LDKC-CU-id*, *internal*, or *virtual* is specified as the `LDEVEachOfCU` operand, data that can be saved when internal or virtual is specified is saved.
Port [[*port-name*]:[*port-name*]][…]	Use this operand when you want to export port statistics, which are displayed in the Monitor Performance window. When statistics are exported in a ZIP file, the file name will be Port_dat.ZIP. For details on the statistics exported by this operand, see Port statistics. When you specify variable *port-name*, you can specify a range of ports for which data is saved using the format *port-name:port-name*. For example, `CL3−a:CL3−c` indicates ports from CL3-a to CL3-c. Ensure that the *port-name* value on the left of the colon is smaller than the *port-name* value on the right of the colon. The smallest *port-name* value is CL1-A and the largest *port-name* value is CL4-r. The following formula illustrates which value is smaller than which value: CL1-A < CL1-B < … < CL2-A < CL2-B < … < CL3-a < CL3-b < … < CL4-a < … < CL4-r For example, you can specify `Port CL1−C:CL2−A`, but you cannot specify `Port CL2−A:CL1−C`. Also, you can specify `Port CL3−a:CL3−c`, but you cannot specify `Port CL3−c:CL3−a`. If *port-name* is not specified, the monitoring data of all ports will be exported.
MFPort [[*port-name*]:[*port-name*]][…] (VSP 5000 series)	Use this operand when you want to save monitoring data for mainframe fibre ports displayed on the Performance Monitor window. When statistics are exported to a ZIP file, the file name is MF_Port_dat.ZIP. For details on the statistics exported by this operand, see Port statistics. When you specify variable *port-name*, you can narrow a range of ports for which data is saved using the format *port-name:port-name*. For example, `CL1−C.01:CL1−C.03` indicates the range from host group #01 of port CL1-C to host group #03 of port CL1-C. Ensure that the value on the left of the colon is smaller than the value on the right of the colon. The smallest *port-name* value is CL1-A and the largest *port-name* value is CL4-r. The following formula illustrates which *port-name* value is smaller than which *port-name* value: CL1-A < CL1-B < … < CL2-A < CL2-B < … < CL3-a < CL3-b < … < CL4-a < …<CL4-r For example, you can specify CL1-C:CL2-A, but you cannot specify CL2-A:CL1-C. Also, you can specify CL3-a:CL3-c, but you cannot specify CL3-c:CL3-a. If *port-name* is not specified, the monitoring data of all volumes is exported.

Operand	Description
PortWWN [[*port-name*]:[*port-name*]][…]	Use this operand when you want to export statistics about host bus adapters (WWNs) connected to ports, which are displayed in the Monitor Performance window. When statistics are exported in a ZIP file, the file name will be PortWWN_dat.ZIP. For details on the statistics exported by this operand, see Port statistics. When you specify variable *port-name*, you can narrow a range of ports for which data is saved using the format *port-name:port-name*. For example, `CL3-a:CL3-c` indicates ports from CL3-a to CL3-c. Ensure that the *port-name* value on the left of the colon is smaller than the *port-name* value on the right of the colon. The smallest *port-name* value is CL1-A and the largest *port-name* value is CL4-r. The following formula illustrates which value is smaller than which value: CL1-A < CL1-B < … < CL2-A < CL2-B < … < CL3-a < CL3-b < … < CL4-a < … < CL4-r For example, you can specify `PortWWN CL1-C:CL2-A`, but you cannot specify `PortWWN CL2-A:CL1-C`. Also, you can specify `PortWWN CL3-a:CL3-c`, but you cannot specify `PortWWN CL3-c:CL3-a`. If *port-name* is not specified, the monitoring data of all host bus adapters will be exported.
LU [[*port-name.host-group-id]*:[*port-name.host-group-id*]][…]	Use this operand when you want to export statistics about LU paths, which are displayed in the Monitor Performance window. When statistics are exported in a ZIP file, the file name will be LU_dat.ZIP. For details on the statistics exported by this operand, see Volumes (LU) statistics. When you specify variable *port-name.host-group-id*, you can narrow the range of LU paths whose monitoring data are to be exported. *port-name* is a port name. *host-group-id* is the ID of a host group (that is, a host storage domain). The host group (host storage domain) ID must be a hexadecimal numeral. The colon (:) indicates a range. For example, `CL1-C.01:CL1-C.03` indicates the range from the host group #01 of the CL1-C port to the host group #03 of the CL1-C port. Ensure that the value on the left of the colon is smaller than the value on the right of the colon. The smallest *port-name* value is CL1-A and the largest *port-name* value is CL4-r. The following formula illustrates which *port-name* value is smaller than which *port-name* value: CL1-A < CL1-B < … < CL2-A < CL2-B < … < CL3-a < CL3-b < … < CL4-a < … < CL4-r For example, you can specify `LU CL1-C.01:CL2-A.01`, but you cannot specify `LU CL2-A.01:CL1-C.01`. Also, you can specify `LU CL1-C.01:CL1-C.03`, but you cannot specify `LU CL1-C.03:CL1-C.01`. If *port-name.host-group-id* is not specified, the monitoring data of all LU paths will be exported.
PPCGWWN [[*monitor-target-*	Use this operand when you want to export statistics about all host bus adapters connected to ports, which are displayed in the Monitor Performance window. When statistics are exported in a ZIP file, the file name will be PPCGWWN_dat.ZIP. For

Operand	Description
name]:[*monitor-target-name*]][…]	details on the statistics exported by this operand, see All host bus adapters connected to ports. When you specify variable *monitor-target-name*, you can narrow the range of monitoring target groups whose monitoring data are to be exported. *monitor-target-name* is the name of an monitoring target group. If the name includes any non-alphanumeric character, the name must be enclosed by double quotation marks ("). The colon (:) indicates a range. For example, `Grp01:Grp03` indicates a range of SPM groups from Grp01 to Grp03. Ensure that the *monitor-target-name* value on the left of the colon is smaller than the *monitor-target-name* value on the right of the colon. Numerals are smaller than letters and lowercase letters are smaller than uppercase letters. In the following formulas, values are arranged so that smaller values are on the left and larger values are on the right: • 0 < 1 < 2 < …< 9 < a < b < …< z < A < B < … < Z • cygnus < raid < Cancer < Pisces < RAID < RAID5 If *monitor-target-name* is not specified, the monitoring data of all host bus adapters will be exported.
RemoteCopy	Use this operand when you want to export statistics about remote copy operations of TrueCopy, TrueCopy for Mainframe, and global-active device. By using this operand, you can export monitoring data about remote copy operations performed by TC, TCz, and GAD in the whole volumes. When statistics are exported to a ZIP file, the file name will be RemoteCopy_dat.ZIP. For details on the statistics exported by this operand, see Remote copy operations by TC/TCz and monitoring data by GAD (whole volumes).
RCLU [[*port-name.host-group-id*]:[*port-name.host-group-id*]][…]	Use this operand when you want to export statistics about remote copy operations of TrueCopy and global-active device. By using this operand, you can export monitoring data about remote copy operations performed by TC and GAD at each volume (LU). When statistics are exported to a ZIP file, the file name will be RCLU_dat.ZIP. (VSP 5000 series) Note that this data cannot be obtained from a mainfame volume. For details on the statistics exported by this operand, see Remote copy operations by TC and monitoring data by GAD (for each volume (LU)). When you specify variable *port-name.host-group-id*, you can narrow the range of LU paths whose monitoring data are to be exported, where *port-name* is a port name and *host-group-id* is the ID of a host group. The host group ID must be a hexadecimal numeral. The colon (:) indicates a range. For example, `CL1-C.01:CL1-C.03` indicates the range from the host group #01 of the CL1-C port to the host group #03 of the CL1-C port. Ensure that the value on the left of the colon is smaller than the value on the right of the colon. The smallest *port-name* value is CL1-A and the largest *port-name* value is CL4-r. The following formula illustrates which *port-name* value is smaller than which *port-name* value: CL1-A < CL1-B < … < CL2-A < CL2-B < … < CL3-a < CL3-b < … < CL4-a < … < CL4-r

Operand	Description
	For example, you can specify RCLU `CL1−C.01:CL2−A.01`, but you cannot specify RCLU `CL2−A.01:CL1−C.01`. Also, you can specify RCLU `CL1−C.01:CL1−C.03`, but you cannot specify RCLU `CL1−C.03:CL1−C.01`. If *port-name.host-group-id* is not specified, the monitoring data of all volumes (LUs) will be exported.
RCLDEV [[*LDKC-CU-id*]:[*LDKC-CU-id*]][…]|	Use this operand when you want to export statistics about remote copy operations of TrueCopy, TrueCopy for Mainframe, and global-active device. By using this operand, you can export monitoring data about remote copy operations performed by TC, TCz, and GAD at volumes controlled by each CU. When statistics are exported to a ZIP file, multiple ZIP files whose names are beginning with `RCLDEV_` will be output. For details on the statistics exported by this operand, see Remote copy by TC/TCz and monitoring data by GAD (volumes controlled by a particular CU). When you specify variable *LDKC-CU-id*, you can narrow the range of LDKC:CUs that control the volumes whose monitoring data are to be exported. *LDKC-CU-id* is an ID of an LDKC:CU. The colon (:) indicates a range. For example, `000:105` indicates LDKC:CUs from 00:00 to 01:05. Ensure that the *LDKC-CU-id* value on the left of the colon is smaller than the *LDKC-CU-id* value on the right of the colon. For example, you can specify RCLDEV `000:105`, but you cannot specify RCLDEV `105:000`. If *LDKC-CU-id* is not specified, the monitoring data of all volumes will be exported.
UniversalReplicator	Use this operand when you want to export statistics about remote copy operations of UR and URz. By using this operand, you can export monitoring data about remote copy operations performed by Universal Replicator and Universal Replicator for Mainframe in the whole volume. When statistics are exported to a ZIP file, the file name will be UniversalReplicator.zip. For details on the statistics exported by this operand, see Remote copy by UR and URz (whole volumes).
URJNL [[*JNL-group-id]*:[*JNL-group-id*]][…]	Use this operand when you want to export statistics about remote copy operations of UR and URz. By using this operand, you can export monitoring data about remote copy operations performed by Universal Replicatorand Universal Replicator for Mainframe at each journal. When statistics are exported to a ZIP file, the file name will be URJNL_dat.ZIP. For details on the statistics exported by this operand, see Remote copy by UR and URz (at journals). When you specify variable *JNL-group-id*, you can narrow the range of journals whose monitoring data are to be exported. *JNL-group-id* is a journal number. The colon (:) indicates a range. For example, `00:05` indicates journals from 00 to 05. Ensure that the *JNL-group-id* value on the left of the colon is smaller than the *JNL-group-id* value on the right of the colon. For example, you can specify URJNL `00:05`, but you cannot specify URJNL `05:00`. If *JNL-group-id* is not specified, the monitoring data of all journal volumes will be exported.

Operand	Description
URLU [[*port-name.host-group-id*]:[*port-name.host-group-id*]][…]	Use this operand when you want to export statistics about remote copy operations of UR. By using this operand, you can export monitoring data about UR remote copy operations at each volume (LU). When statistics are exported to a ZIP file, the file name will be URLU_dat.ZIP. (VSP 5000 series) Note that this data cannot be obtained from a mainfame volume. For details on the statistics exported by this operand, see [Remote copy by UR (for each volume (LU))](). When you specify variable *port-name.host-group-id*, you can narrow the range of LU paths whose monitoring data are to be exported, where *port-name* is a port name and *host-group-id* is the ID of a host group. The host group ID must be a hexadecimal numeral. The colon (:) indicates a range. For example, `CL1-C.01:CL1-C.03` indicates the range from the host group #01 of the CL1-C port to the host group #03 of the CL1-C port. Ensure that the value on the left of the colon is smaller than the value on the right of the colon. The smallest *port-name* value is CL1-A and the largest *port-name* value is CL4-r. The following formula illustrates which *port-name* value is smaller than which *port-name* value: CL1-A < CL1-B < … < CL2-A < CL2-B < … < CL3-a < CL3-b < … < CL4-a < … < CL4-r For example, you can specify `URLU CL1-C.01:CL2-A.01`, but you cannot specify `URLU CL2-A.01:CL1-C.01`. Also, you can specify `URLU CL1-C.01:CL1-C.03`, but you cannot specify `URLU CL1-C.03:CL1-C.01`. If *port-name.host-group-id* is not specified, the monitoring data of all volumes (LUs) will be exported.
URLDEV [[*LDKC-CU-id*]:[*LDKC-CU-id*]][…]	Use this operand when you want to export statistics about remote copy operations which of UR and URz. By using this operand, you can export monitoring data about remote copy operations performed by Universal Replicator and Universal Replicator for Mainframe at volumes controlled by each CU. When statistics are exported to a ZIP file, multiple ZIP files whose names are beginning with `URLDEV_` will be output. For details on the statistics exported by this operand, see [Remote copy by UR and URz (at volumes controlled by a particular CU)](). When you specify variable *LDKC-CU-id*, you can narrow the range of LDKC:CUs that control the volumes whose monitoring data are to be exported. *LDKC-CU-id* is an ID of an LDKC:CU. The colon (:) indicates a range. For example, `000:105` indicates LDKC:CUs from 00:00 to 01:05. Ensure that the *LDKC-CU-id* value on the left of the colon is smaller than the *LDKC-CU-id* value on the right of the colon. For example, you can specify `URLDEV 000:105`, but you cannot specify `URLDEV 105:000`. If *LDKC-CU-id* is not specified, the monitoring data of all volumes will be exported.

Operand	Description
PhyMPPK	Use this operand when you want to export statistics about MP usage rate of each resource allocated to MP unit in short range. When statistics are exported to a ZIP file, the filename is PHY_MPPK.ZIP. For details on the statistics exported by this operand, see MP units.

Examples

The following example exports statistics about host bus adapters:

```
group PortWWN
```

The following example exports statistics about three ports (CL1-A, CL1-B, and CL1-C):

```
group Port CL1-A:CL1-C
```

The following example exports statistics about six ports (CL1-A to CL1-C, and CL2-A to CL2-C)

```
group Port CL1-A:CL1-C CL2-A:CL2-C
```

The following example exports statistics about the parity group 1-3:

```
group PG 1-3:1-3
```

The following example exports statistics about the parity group 1-3 and other parity groups whose ID is larger than 1-3 (for example, 1-4 and 1-5):

```
group PG 1-3:
```

The following example exports statistics about the external volume groups E1-1 to E1-5:

```
group PG E1-1:E1-5
```

The following example exports statistics about the parity group 1-3 and other parity groups whose ID is smaller than 1-3 (for example, 1-1 and 1-2):

```
group LDEV:1-3
```

The following example exports statistics about LU paths for the host group (host storage domain) ID 01 for the port CL1-A:

```
group LU CL1-A.01:CL1-A.01
```

shortrange (VSP 5000 series)

Description

Use this subcommand to specify a term of monitoring data to be exported into files. Use this subcommand when you want to narrow the export-target term within the stored data.

The shortrange subcommand is valid for monitoring data in short range. Short-range monitoring data appears in the Monitor Performance window when Short-Range is selected as the storing period.

All the monitoring items are stored in short range. Therefore, you can use the shortrange subcommand whichever operand you specify to the group subcommand. If you run Export Tool without specifying the shortrange subcommand, the data stored in the whole monitoring term will be exported.

The login subcommand must run before the shortrange subcommand runs.

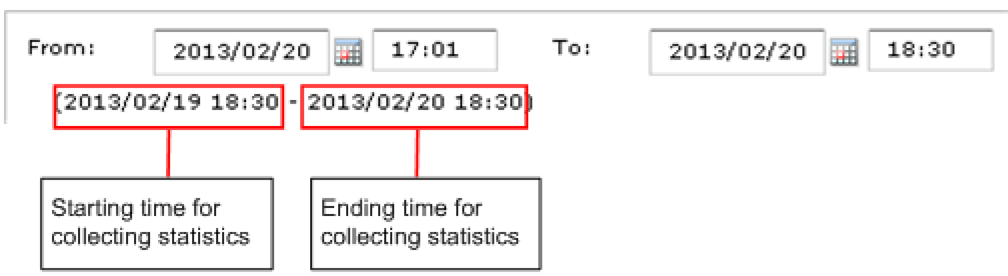
Syntax

shortrange [[*yyyyMMddhhmm*][{+|-}*hhmm*]:[*yyyyMMddhhmm*][{+|-}*hhmm*]]

Operands

The value on the left of the colon (:) specifies the starting time of the period. The value on the right of the colon specifies the ending time of the period. Specify the term within "`Short Range From` XXX `To` XXX" which is output by the show subcommand.

If no value is specified on the left of the colon, the starting time for collecting monitoring data is assumed. If no value is specified on the right of the colon, the ending time for collecting monitoring data is assumed. The starting and ending times for collecting monitoring data are displayed in the Monitoring Term area in the Monitor Performance window.



Operand	Description
yyyyMMddhhmm	*yyyyMMdd* indicates the year, the month, and the day. *hhmm* indicates the hour and the minute. If *yyyyMMddhhmm* is omitted on the left of the colon, the starting time for collecting monitoring data is assumed. If *yyyyMMddhhmm* is omitted on the right of the colon, the ending time for collecting monitoring data is assumed. To save monitoring data up to the sampling end time, omit *yyyyMMddhhmm* on the right of the colon. If you specify *yyyyMMddhhmm* on the right of the colon, specify date and time at least 30 minutes earlier than the current time. Otherwise, an Out of range error may occur.
+hhmm	Adds time (*hhmm*) to *yyyyMMddhhmm* if *yyyyMMddhhmm* is specified. For example, 201306230000+0130 indicates June 23, 2013. 01:30. Adds time to the starting time for collecting monitoring data, if *yyyyMMddhhmm* is omitted.
-hhmm	Subtracts time (hhmm) from *yyyyMMddhhmm* if *yyyyMMddhhmm* is specified. For example, `201306230000-0130` indicates June 22, 2013. 22:30. Subtracts time from the ending time for collecting monitoring data, if *yyyyMMddhhmm* is omitted.

Operand	Description
	If the last two digits of the time on the left or right of the colon (:) are not a multiple of the sampling interval, the time will automatically be changed so that the last two digits is a multiple of the sampling interval. If this change occurs to the time on the left of the colon, the time will be smaller than the original time. If this change occurs to the time on the right of the colon, the time will be larger than the original time. The following are the examples: • If the time on the left is 10:15, the time on the right is 20:30, and the sampling interval is 10 minutes: The time on the left will be changed to 10:10 because the last two digits of the time are not a multiple of 10 minutes. The time on the right will remain unchanged because the last two digits of the time are a multiple of 10 minutes. • If the time on the left is 10:15, the time on the right is 20:30, and the sampling interval is 7 minutes: The time on the left will be changed to 10:14 because the last two digits of the time are not a multiple of 7 minutes. The time on the right will be changed to 20:35 because of the same reason.

Examples

The examples below assume that the:

- Starting time for collecting monitoring data is Jan. 1, 2013, 00:00,
- Ending time for collecting monitoring data is Jan. 2, 2013, 00:00.

`shortrange 201301010930:201301011730`

Export Tool saves monitoring data within the range of Jan. 1, 9:30-17:30.

`shortrange 201301010930:`

Export Tool saves monitoring data within the range of Jan. 1, 9:30 to Jan. 2, 00:00.

`shortrange:201301011730`

Export Tool saves monitoring data within the range of Jan. 1, 0:00-17:30.

`shortrange +0001:`

Export Tool saves monitoring data within the range of Jan. 1, 0:01 to Jan. 2, 00:00.

`shortrange -0001:`

Export Tool saves monitoring data within the range of Jan. 1, 23:59 to Jan. 2, 00:00.

`shortrange:+0001`

Export Tool saves monitoring data within the range of Jan. 1, 0:00-00:01.

`shortrange:-0001`

Export Tool saves monitoring data within the range of Jan. 1, 0:00-23:59.

`shortrange +0101:-0101`

Export Tool saves monitoring data within the range of Jan. 1, 1:01-22:59.

```
shortrange 201301010900+0130:201301011700-0130
```

Export Tool saves monitoring data within the range of Jan. 1, 10:30-15:30.

```
shortrange 201301010900-0130:201301011700+0130
```

Export Tool saves monitoring data within the range of Jan. 1, 7:30-18:30.

```
shortrange 201301010900-0130:
```

Export Tool saves monitoring data within the range of Jan. 1, 7:30 to Jan. 2, 00:00.

longrange (VSP 5000 series)

Description

The longrange subcommand is used to specify a monitoring term (time range) for collecting monitoring data to be exported into files. Use this subcommand when you want to narrow the export-target term within the stored data.

The longrange subcommand is valid for monitoring data in long range. The monitoring data in long range is the contents displayed on the Physical tab of the Performance Management window with selecting longrange as the storing period.

The monitoring items whose data can be stored in long range are limited. The following table shows the monitoring items to which the longrange subcommand can be applied, and also shows the operands to export those monitoring items.

Monitoring Data	Operands of the group subcommand
Usage statistics about parity groups	PhyPG Long
Usage statistics about volumes	PhyLDEV Long
Usage statistics about MPs and data recovery and reconstruction processors	PhyProc Long
Usage statistics about access paths and write pending rate	PhyMPU Long

If you run Export Tool without specifying the longrange subcommand, the data stored in the whole monitoring term will be exported.

The login subcommand must run before the longrange subcommand runs.

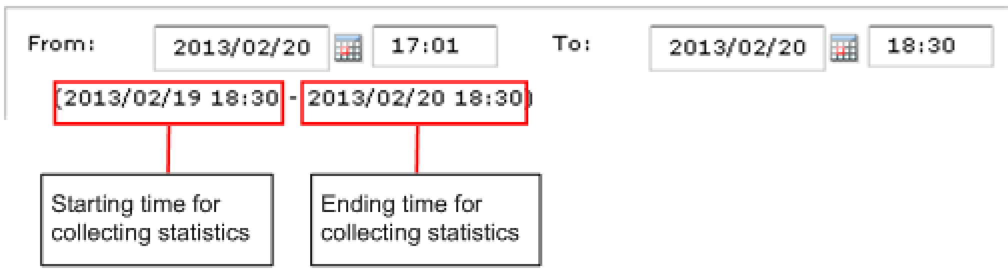
Syntax

longrange [[*yyyyMMddhhmm*][{+|-}*ddhhmm*]:[*yyyyMMddhhmm*][{+|-}*ddhhmm*]]

Operands

The value on the left of the colon (:) specifies the starting time of the period. The value on the right of the colon specifies the ending time of the period. Specify the term within "`Long Range From` XXX `To` XXX" which is output by the show subcommand.

If no value is specified on the left of the colon, the starting time for collecting monitoring data is assumed. If no value is specified on the right of the colon, the ending time for collecting monitoring data is assumed. The starting and ending times for collecting monitoring data are displayed in the Monitoring Term area in the Monitor Performance window.



Operand	Description
yyyyMMddhhmm	*yyyyMMdd* indicates the year, the month, and the day. *hhmm* indicates the hour and the minute. If *yyyyMMddhhmm* is omitted on the left of the colon, the starting time for collecting monitoring data is assumed. If *yyyyMMddhhmm* is omitted on the right of the colon, the ending time for collecting monitoring data is assumed.
+ddhhmm	Adds time (*ddhhmm*) to *yyyyMMddhhmm* if *yyyyMMddhhmm* is specified. For example, `201301120000+010130` indicates Jan. 13, 2013. 01:30. Adds time to the starting time for collecting monitoring data, if *yyyyMMddhhmm* is omitted.
-ddhhmm	Subtracts time (ddhhmm) from *yyyyMMddhhmm* if *yyyyMMddhhmm* is specified. For example, `201301120000-010130` indicates Jan. 10, 2013. 22:30. Subtracts time from the ending time for collecting monitoring data, if *yyyyMMddhhmm* is omitted. Ensure that *mm* is 00, 15, 30, or 45. If you do not specify *mm* in this way, the value on the left of the colon (:) will be rounded down to one of the four values. Also, the value on the right of the colon will be rounded up to one of the four values. For example, if you specify `201301010013:201301010048`, the specified value is regarded as `201301010000:201301010100`.

Examples

The examples below assume that:

- the starting time for collecting monitoring data is Jan. 1, 2013, 00:00.
- the ending time for collecting monitoring data is Jan. 2, 2013, 00:00.

```
longrange 201301010930:201301011730
```

Export Tool saves monitoring data within the range of Jan. 1, 9:30-17:30.

`longrange 201301010930:`

Export Tool saves monitoring data within the range of Jan. 1, 9:30 to Jan. 2, 00:00.

`longrange:201301011730`

Export Tool saves monitoring data within the range of Jan. 1, 0:00-17:30.

`longrange +000015:`

Export Tool saves monitoring data within the range of Jan. 1, 0:15 to Jan. 2, 00:00.

`longrange -000015:`

Export Tool saves monitoring data within the range of Jan. 1, 23:45 to Jan. 2, 00:00.

`longrange:+000015`

Export Tool saves monitoring data within the range of Jan. 1, 0:00-00:15.

`longrange:-000015`

Export Tool saves monitoring data within the range of Jan. 1, 0:00-23:45.

`longrange +000115:-000115`

Export Tool saves monitoring data within the range of Jan. 1, 1:15-22:45.

`longrange 201301010900+000130:201301011700-000130`

Export Tool saves monitoring data within the range of Jan. 1, 10:30-15:30.

`longrange 201301010900-000130:201301011700+000130`

Export Tool saves monitoring data within the range of Jan. 1, 7:30-18:30.

`longrange 201301010900-000130:`

Export Tool saves monitoring data within the range of Jan. 1, 7:30 to Jan. 2, 00:00.

range

Description

(VSP E series) Use this subcommand to specify a term of monitoring data to be exported into files. Use this subcommand when you want to narrow the export-target term within the stored data.

You need to log into the storage system with the login subcommand before the range subcommand runs.

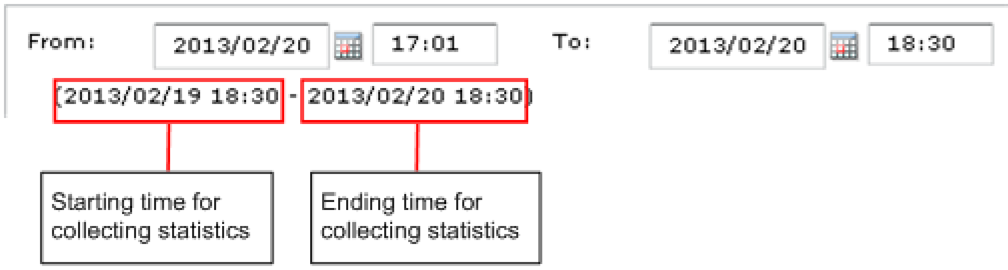
Syntax

range [[*yyyyMMddhhmm*][{+|-}*hhmm*]:[*yyyyMMddhhmm*][{+|-}*hhmm*]]

Operands

The value on the left of the colon (:) specifies the starting time of the period. The value on the right of the colon specifies the ending time of the period. Specify the term within "`Range From` XXX `To` XXX" which is output by the show subcommand.

If no value is specified on the left of the colon, the starting time for collecting monitoring data is assumed. If no value is specified on the right of the colon, the ending time for collecting monitoring data is assumed. The starting and ending times for collecting monitoring data are displayed in the Monitoring Term area in the Monitor Performance window.



Operand	Description
yyyyMMddhhmm	*yyyyMMdd* indicates the year, the month, and the day. *hhmm* indicates the hour and the minute. If *yyyyMMddhhmm* is omitted on the left of the colon, the starting time for collecting monitoring data is assumed. If *yyyyMMddhhmm* is omitted on the right of the colon, the ending time for collecting monitoring data is assumed. To save monitoring data up to the sampling end time, omit *yyyyMMddhhmm* on the right of the colon. If you specify *yyyyMMddhhmm* on the right of the colon, specify date and time at least 30 minutes earlier than the current time. Otherwise, an Out of range error may occur.
+hhmm	Adds time (*hhmm*) to *yyyyMMddhhmm* if *yyyyMMddhhmm* is specified. For example, 201306230000+0130 indicates June 23, 2013. 01:30. Adds time to the starting time for collecting monitoring data, if *yyyyMMddhhmm* is omitted.
-hhmm	Subtracts time (*hhmm*) from *yyyyMMddhhmm* if *yyyyMMddhhmm* is specified. For example, 201306230000-0130 indicates June 22, 2013. 22:30. Subtracts time from the ending time for collecting monitoring data, if *yyyyMMddhhmm* is omitted. If the last two digit of the time on the left or right of the colon (:) is not a multiple of the sampling interval, the time will automatically be changed so that the last two digits is a multiple of the sampling interval. If this change occurs to the time on the left of the colon, the time will be smaller than the original time. If this change occurs to the time on the right of the colon, the time will be larger than the original time. The following are the examples: • If the time on the left is 10:15, the time on the right is 20:30, and the sampling interval is 10 minutes: The time on the left will be changed to 10:10 because the last two digits of the time is not a multiple of 10 minutes. The time on the right will remain unchanged because the last two digits of the time is a multiple of 10 minutes

Operand	Description
	• If the time on the left is 10:15, the time on the right is 20:30, and the sampling interval is 7 minutes: The time on the left will be changed to 10:14 because the last two digits of the time is not a multiple of 7 minutes. The time on the right will be changed to 20:35 because of the same reason.

Examples

The following examples assume that:

- The starting time for collecting monitoring data is Jan. 1, 2013, 00:00.
- The ending time for collecting monitoring data is Jan. 2, 2013, 00:00.

```
range 201301010930:201301011730
```

Export Tool saves monitoring data within the range of Jan. 1, 9:30-17:30.

```
range 201301010930:
```

Export Tool saves monitoring data within the range of Jan. 1, 9:30 to Jan. 2, 00:00.

```
range:201301011730
```

Export Tool saves monitoring data within the range of Jan. 1, 0:00-17:30.

```
range +0001:
```

Export Tool saves monitoring data within the range of Jan. 1, 0:01 to Jan. 2, 00:00.

```
range :-0001
```

Export Tool saves monitoring data within the range of Jan. 1, 0:00-23:59.

```
range +0101:-0101
```

Export Tool saves monitoring data within the range of Jan. 1, 1:01-22:59.

```
range 201301010900+0130:201301011700-0130
```

Export Tool saves monitoring data within the range of Jan. 1, 10:30-15:30.

```
range 201301010900-0130:201301011700+0130
```

Export Tool saves monitoring data within the range of Jan. 1, 7:30-18:30.

```
range 201301010900-0130:
```

Export Tool saves monitoring data within the range of Jan. 1, 7:30 to Jan. 2, 00:00.

outpath

Description

The outpath subcommand specifies the directory to which monitoring data will be exported.

Syntax

outpath [*path*]

Operands

Operand	Description
path	Specifies the directory in which files will be saved. If the directory includes any non-alphanumeric character, the directory must be enclosed by double quotation marks ("). Use a double back slash (`\\`) as the directory path delimiter. If the specified directory does not exist, this subcommand creates a directory that has the specified name. If this operand is omitted, the current directory is assumed.

Examples

The following example saves files in the directory C:\Project\out on a Windows computer:

```
outpath "C:\\Project\\out"
```

The following example saves files in the out directory in the current directory:

```
outpath out
```

option

Description

This subcommand specifies the following:

- whether to compress monitoring data in ZIP files
- whether to overwrite or delete existing files and directories when saving monitoring data in files

Syntax

option [`compress`|`nocompress`] [`ask`|`clear`|`noclear`]

Operands

Operand	Description
The two operands below specify whether to compress CSV files into ZIP files. If none of these operands is specified, compress is assumed.	

Operand	Description
compress	Compresses data in ZIP files. To extract CSV files out of a ZIP file, you will need to decompress the ZIP file.
nocompress	Does not compress data in ZIP files and saves data in CSV files.
The three operands below specify whether to overwrite or delete an existing file or directory when Export Tool saves files. If none of these operands is specified, `ask` is assumed.	
ask	Displays a message that asks whether to delete existing files or directories.
clear	Deletes existing files and directories and then saves monitoring data in files.
noclear	Overwrites existing files and directories.

Example

The following example saves monitoring data in CSV files, not in ZIP files:

```
option nocompress
```

apply

Description

The apply subcommand saves monitoring data specified by the group subcommand into files.

(VSP 5000 series) The login subcommand must run before the apply subcommand runs.

(VSP E series) You need to log into the storage system with the login subcommand before the apply subcommand runs.

The apply subcommand does nothing if the group subcommand is running.

The settings made by the group subcommand will be reset when the apply subcommand finishes.

Syntax

apply

set

Description

The set subcommand starts or ends monitoring the storage system (for example, starting or ending the collection of performance statistics). The set subcommand also specifies the sampling interval (interval of collecting statistics).

If you want to use the set subcommand, you must use the login subcommand (see login to log in to the storage system). Ensure that the set subcommand runs immediately before Export Tool finishes.

Executing the set subcommand generates an error in the following conditions:

- Some other user is being logged onto SVP in Modify mode.
- Maintenance operations are being performed at SVP.

If an error occurs, do the following:

- Check with Device Manager - Storage Navigator and so on, and confirm that all users who are logged onto the storage system are not in Modify mode. If any user is logged on in Modify mode, ask the user to switch to View mode.
- Wait until maintenance operations finish at SVP, so that the set subcommand can run.
  Note: Following are notes of the set command.
  - Batch files can include script that will run when an error occurs.
  - When the set subcommand starts or ends the monitoring or changes the sampling interval after the Monitor Performance window is started, the contents displayed in the Monitor Performance window do not change automatically in conjunction with the set subcommand operation. To display the current monitoring status in the Monitor Performance window, click Refresh on the menu bar of the Device Manager - Storage Navigator main window.
  - If you change the specified sampling interval during a monitoring, the previously collected monitoring data will be deleted.

Syntax

set [`switch=`{*m*|`off`}]

Operands

Operand	Description
switch={*m*|off}	To start monitoring, specify the sampling interval (interval of collecting statistics) of monitoring data at *m*. Specify a value between 1 and 15 in minutes. (VSP E series) *m* is the sampling interval by Performance Monitor. (VSP 5000 series) The sampling interval in long range is fixed to 15 minutes. m is the sampling interval in short range monitoring by Performance Monitor. To end monitoring, specify `off`. If this operand is omitted, the set subcommand does not make settings for starting or ending monitoring.

Examples

The following command file saves port statistics and then ends monitoring ports:

(VSP 5000 series)

```
svpip 158.214.135.57
login expusr passwd
show
group Port
```

```
short-range 201304010850:201304010910
apply
set switch=off
```

(VSP E series)

```
ip 158.214.135.57
dkcsn 123456
login expusr passwd
show
group Port
range 200604010850:200604010910
apply
set switch=off
```

The following command file starts monitoring remote copy operations. The sampling time interval is 10 minutes:

(VSP 5000 series)

```
svpip 158.214.135.57 login expusr passwd set switch=10
```

(VSP E series)

```
ip 158.214.135.57
dkcsn 123456
login expusr passwd
set switch=10
```

help

Description

The help subcommand displays the online help for subcommands.

If you want to view the online help, you should create a batch file and a command file that are exclusively used for displaying the online help. For detailed information, see the following Example.

Syntax

help

Example

In this example, a command file (cmdHelp.txt) and a batch file (runHelp.bat) are created in the C:\export directory on a Windows computer:

- Command file (c:\export\cmdHelp.txt):

```
help
```

- Batch file (c:\export\runHelp.bat):

```
java -classpath "./lib/JSanExportLoader.jar"
-Del.tool.Xmx=536870912 -Dmd.command=cmdHelp.txt
-Dmd.logpath=log sanproject.getexptool.RJElMain<CR+LF>
pause<CR+LF>
```

In the preceding script, <CR+LF> indicates the end of a command line.

In this example, you must do one of the following to view the online help:

- Double-click runHelp.bat.
- Go to the c:\export directory at the command prompt, enter `runHelp` or `runHelp.bat`, and then press Enter.

Java

Description

This command starts Export Tool and exports monitoring data into files. To start Export Tool, write this Java command in your batch file and then run the batch file.

Syntax

Java-classpath*class-pathrproperty-parameters*sanproject.getexptool.RJElMain

Operands

Operand	Description
class-path	Specifies the path to the class file of Export Tool. The path must be enclosed in double quotation marks (").
property-parameters	You can specify the following parameters. At minimum you must specify -Dmd.command. • -Dhttp.proxyHost= *host-name-of-proxy-host*, or -Dhttp.proxyHost=*IP-address-of-proxy-host* Specifies the host name or the IP address of a proxy host. You must specify this parameter if the computer that runs Export Tool communicates with SVP through a proxy host. • -Dhttp.proxyPort=*port-number-of-proxy-host* Specifies the port number of a proxy host. You must specify this parameter if the computer that runs Export Tool communicates with SVP through a proxy host. • -Del.tool.Xmx=*VM-heap-size-when-ExportTool-starts* (bytes) Specifies the size of memory to be used by JRE when Export Tool is being run. You must specify this parameter. The memory size must be 536870912, as shown in the Example below. If an installed memory size is smaller than the recommended size of the PC running Device Manager - Storage Navigator, you must install more memory before executing Export Tool. If an installed memory size is larger than the recommended memory size of the PC running Device Manager - Storage Navigator, you can specify a memory size larger than as shown in the Example. However, to prevent lowering of execution speed, you do not set oversized memory size.

Operand	Description
	• -Dmd.command=*path-to-command-file* Specifies the path to the command file • -Dmd.logpath=*path-to-log-file* Specifies the path to log files. A log file will be created whenever Export Tool runs. If this parameter is omitted, log files will be saved in the current directory. • -Dmd.logfile=*name-of-log-file* Specifies the name of the log file. If this parameter is omitted, log files are named exportMMddHHmmss.log. MMddHHmmss indicates when Export Tool runs. For example, the log file export0101091010.log contains log information about the Export Tool execution at Jan. 1, 09:10:10. • -Dmd.rmitimeout=*timeout(min.)* Specifies the timeout value for communication between Export Tool and the SVP: ◦ Default: 20 minutes ◦ Minimum: 1 minute ◦ Maximum: 1,440 minutes (24 hours) If a request does not come from Export Tool within the timeout period, the SVP determines that execution has stopped and disconnects the session with Export Tool. Therefore, if the machine on which Export Tool is running is slow, Export Tool sessions may be disconnected unexpectedly. To prevent this from occurring, increase the timeout period by entering a larger value in this parameter. • -Del.logpath=*log-output-directory-name* Specify the directory to store the log file generated when Export Tool was downloaded. By default, the current directory is set. The initial value for the startup batch file is log. • -Del.logfile=*log-file-name* Specify the name of the log file generated when Export Tool was downloaded. By default, loaderMMddHHmmss.log is set. *MM* is the month, *dd* is the date, and *HH* is the hour, *mm* is minutes, and *ss* is seconds. When the default name is used, a new log file is created every time you run Export Tool. As a result, you need to delete log files regularly. The initial value for the startup batch file is not specified. • -Del.mode=*startup-mode-of-Export-Tool* (all/delete) Specify the startup mode of Export Tool. The following shows startup modes and their behavior. ◦ Mode: all

<table>
<tr><th>Operand</th><th>Description</th></tr>
<tr><td></td><td>
Download Export Tool: Yes
Run Export Tool: Yes
Delete temporary directories in the lib directory: Yes
Mode: delete
Download Export Tool: no
Run Export Tool: no
Delete temporary directories in the lib directory: Yes
(VSP E series) -Del.dlport=port-number-used-when-Export-Tool-is-downloaded
Specifies the port number used when Export Tool is downloaded. By default, 51099 is set. When you change the port number (port number key name: RMIClassLoader) of the SVP, you must also change the port number of this operand.
For settings that are affected by changes of port numbers for the SVP, see the System Administrator Guide.
(VSP 5000 series) -Del.security={ignore|verify}
Specifies the settings to disable or enable use of the server verification for the SSL communications.
ignore: Specifying ignore disables the server verification.
The server verification can be performed, however, even if the verification cannot be performed, the Export Tool processing will not be suspended. ignore is set by default.
verify: Specifying verify enables the server verification.
If this operand is omitted, ignore will be set. If you specify a character string other than ignore and verify, the following message appears and the Export Tool processing will be suspended: Invalid value : "el.security = nn"
</td></tr>
</table>

Examples

The following example assumes that the computer running Export Tool communicates with the SVP through a proxy host. In the following example, the host name of the proxy host is Jupiter, and the port name of the proxy host is 8080:

```
java -classpath "./lib/JSanExportLoader.jar"
-Dhttp.proxyHost=Jupiter -Dhttp.proxyPort=8080 -Del.tool.Xmx=536870912
-Dmd.command=command.txt
-Dmd.rmitimeout=20
-Dmd.logpath=log sanproject.getexptool.RJElMain <CR+LF>
```

In the following example, a log file named export.log will be created in the log directory below the current directory when Export Tool runs:

```
java -classpath "./lib/JSanExportLoader.jar"
-Del.tool.Xmx=536870912 -Dmd.command=command.txt -Dmd.logfile=export.log
-Dmd.logpath=log sanproject.getexptool.RJElMain<CR+LF>
```

In the above script, <CR+LF> indicates the end of a command line.

Viewing exported files using Export Tool

Export Tool saves the exported monitoring data into text files in CSV (comma-separated value) format, in which values are delimited by commas. Many spreadsheet applications can be used to open CSV files.

Export Tool by default saves the CSV text files in compressed (ZIP) files. To use a text editor or spreadsheet software to view or edit the monitoring data, first decompress the ZIP files to extract the CSV files. You can also configure Export Tool to save monitoring data in CSV files instead of ZIP files.

Monitoring data exported by Export Tool

The following table shows the correspondence between the Performance Management windows and the monitoring data that can be exported by Export Tool. For details on the names of the ZIP files and CSV files used for saving monitoring data, see the tables indicated in the links in the Monitoring data column.

For details about the monitoring data to be saved in files, see the table under the applicable topic listed in the Monitoring data column in the following table. If the Monitor Performance window displays an instantaneous value of monitoring data at the time the data is collected, the applicable monitoring data name is indicated as such by a footnote. For monitoring data that is not specified to show an instantaneous value, the Monitor Performance window displays the average value of data over the sampling interval. The following sampling intervals can be set in the Edit Monitoring Switch window :

- (VSP 5000 series) The sampling intervals are 1 to 15 minutes and 15 minutes for Short Range and Long Range, respectively.
- (VSP E series) The sampling intervals are 1 to 15 minutes.

GUI operation	Monitoring data
Select Parity Groups from Object list in Performance Objects field in Monitor Performance window.	Resource usage and write-pending rate statistics Parity group and external volume group statistics
Select Logical Devices (Base) from Object list in Performance Objects field in Monitor Performance window.	Resource usage and write-pending rate statistics Statistics for volumes in parity/external volume groups Volumes in parity groups or external volume groups (at volumes controlled by a particular CU)
Select Logical Devices (TC/TCz/GAD) from Object list in Performance Objects field in Monitor Performance window.	Remote copy by TC/TCz and monitoring data by GAD (volumes controlled by a particular CU)

GUI operation	Monitoring data
Select Logical Devices (UR/URz) from Object list in Performance Objects field in Monitor Performance window.	Remote copy by UR and URz (at volumes controlled by a particular CU)
Select Access Path from Object list in Performance Objects field in Monitor Performance window. (VSP 5000 series)	Resource usage and write-pending rate statistics
Select Cache from Object list in Performance Objects field in Monitor Performance window.	Resource usage and write-pending rate statistics
Select Controller from Object list in Performance Objects field in Monitor Performance window.	Resource usage and write-pending rate statistics MP units (VSP 5000 series)
Select Fibre Port from Object list in Performance Objects field in Monitor Performance window.	Port statistics
Select iSCSI Port from Object list in Performance Objects field in Monitor Performance window.	Port statistics
Select Mainframe Fibre Port from Object list in Performance Objects field in Monitor Performance window. (VSP 5000 series)	Port statistics
Select LUN Base from Object list in Performance Objects field in Monitor Performance window.	Volumes (LU) statistics
Select LUN (TC/GAD) from Object list in Performance Objects field in Monitor Performance window.	Remote copy operations by TC and monitoring data by GAD (for each volume (LU))
Select LUN (UR/URz) from Object list in Performance Objects field in Monitor Performance window.	Remote copy by UR (for each volume (LU))
Select WWN from Object list in Performance Objects field in Monitor Performance window.	Host bus adapters connected to ports statistics All host bus adapters connected to ports

GUI operation	Monitoring data
Select Journal from Object list in Performance Objects field in Monitor Performance window.	Remote copy by UR and URz (at journals)
Select Entire Storage System (TC/TCz/GAD) from Object list in Performance Objects field in Monitor Performance window.	Remote copy operations by TC/TCz and monitoring data by GAD (whole volumes)
Select Entire Storage System (UR/URz) from Object list in Performance Objects field in Monitor Performance window.	Remote copy by UR and URz (whole volumes)

Resource usage and write-pending rate statistics

The following table shows the file names and types of information in the Monitor Performance window that can be saved to files using Export Tool. These files contain statistics about resource usage and write pending rates.

ZIP file	CSV file	Data saved in the file
PhyPG_dat.ZIP	PHY_Long_PG.csv and PHY_Short_PG.csv (VSP 5000 series) PHY_PG.csv (VSP E series)	Usage rates for parity groups in long range and short range, respectively (%). Usage rates for parity groups.
PhyLDEV_dat.ZIP	PHY_Long_LDEV_x-y.csv and PHY_Short_LDEV_x-y.csv (VSP 5000 series) PHY_LDEV_x-y.csv (VSP E series)	Usage rates for volumes in a parity group in long range and short range, respectively (%). Usage rates for volumes in a parity group

ZIP file	CSV file	Data saved in the file
	PHY_Short_LDEV_SI_x-y.csv (VSP 5000 series) PHY_LDEV_SI_xy.csv (VSP E series)	Usage rates for ShadowImage and ShadowImage for Mainframe volumes in a parity group in short range (%). Usage rates for ShadowImage volumes in a parity group
PhyExG_dat.ZIP	PHY_ExG_Response.csv	This file includes the average response time for the volume groups including external storage volumes (milliseconds).
	PHY_ExG_Trans.csv	This file includes the amount of transferred data for volume groups including external storage volumes (KB/s).
	PHY_ExG_Read_Response.csv	This file includes the average read response time for the volume

ZIP file	CSV file	Data saved in the file
		groups including external storage volumes (milliseconds).
	PHY_ExG_Write_Response.csv	This file includes the average write response time for the volume groups including external storage volumes (milliseconds).
	PHY_ExG_Read_Trans.csv	This file includes the amount of read transferred data for volume groups including external storage volumes (KB/s).
	PHY_ExG_Write_Trans.csv	This file includes the amount of write transferred data for volume groups including external storage volumes (KB/s).

ZIP file	CSV file	Data saved in the file
PhyExLDEV_dat/PHY_ExLDEV_Response.ZIP	PHY_ExLDEV_Response_x-y.csv	This file includes the average response time for external storage volumes in the volume group x-y (milliseconds).
PhyExLDEV_dat/PHY_ExLDEV_Trans.ZIP	PHY_ExLDEV_Trans_x-y.csv	This file includes the amount of data transferred for external storage volumes in the volume group x-y (KB/s).
PhyExLDEV_dat/PHY_ExLDEV_Read_Response.ZIP	PHY_ExLDEV_Read_Response_x-y.csv	This file includes the average reading response time for external storage volumes in the volume group x-y (milliseconds).
PhyExLDEV_dat/PHY_ExLDEV_Write_Response.ZIP	PHY_ExLDEV_Write_Response_x-y.csv	This file includes the average writing response time for external storage volumes in the volume group x-y (milliseconds).

ZIP file	CSV file	Data saved in the file
PhyExLDEV_dat/PHY_ExLDEV_Read_Trans.ZIP	PHY_ExLDEV_Read_Trans_x-y.csv	This file includes the amount of reading data transferred for external storage volumes in the volume group x-y (KB/s).
PhyExLDEV_dat/PHY_ExLDEV_Write_Trans.ZIP	PHY_ExLDEV_Write_Trans_x-y.csv	This file includes the amount of writing data transferred for external storage volumes in the volume group x-y (KB/s).
PhyProc_dat.ZIP (VSP 5000 series)	PHY_Long_MP.csv	Usage rates for MPs in long range (%).
	PHY_Short_MP.csv	Usage rates for MPs in short range (%).
	PHY_Long_DRR.csv	Usage rates for DRRs (data recovery and reconstruction processors) in long range.
	PHY_Short_DRR.csv	Usage rates for DRRs (data recovery

ZIP file	CSV file	Data saved in the file
		and reconstruction processors) in short range.
PhyProc_dat.ZIP (VSP E series)	PHY_MP.csv	Usage rates for MPs.
PhyMPU_dat.ZIP (VSP 5000 series)	PHY_Long_Write_Pending_Rate.csv	Write pending rates in long range in the entire system.
	PHY_Short_Write_Pending_Rate.csv	Write pending rates in short range in the entire system.*
	PHY_Short_Cache_Usage_Rate.csv	Usage rates for cache memory in the entire system.*
	PHY_Long_Write_Pending_Rate_z.csv	Write pending rates in long range in each MP unit.
	PHY_Short_Write_Pending_Rate_z.csv	Write pending rates in short range in each MP unit.*
	PHY_Short_Cache_Usage_Rate_z.csv	Usage rates for cache memory in each MP unit.*
	PHY_Cache_Allocate_z.csv	The allocated size of the cache memory in each MP unit (MB).* This value does not correspond to the total capacity of

ZIP file	CSV file	Data saved in the file
		cache because the value is the same as the allocated size of the cache memory that is managed by an MP unit.
	PHY_Long_HIE_ISW.csv	Usage rates of access path between HIE s and ISWs in long range.
	PHY_Long_MPU_HIE.csv	Usage rates of access path between MP units and HIEs in long range.
	PHY_Short_HIE_ISW.csv	Usage rates of access path between HIEs and ISWs in short range.*
	PHY_Short_MPU_HIE.csv	Usage rates of access path between MP units and HIEs in short range.*
PhyMPU_dat.ZIP (VSP E series)	PHY_Cache_Usage_Rate.csv	Usage rates for cache memory in the entire system.*
	PHY_Write_Pending_Rate.csv	Write pending rates in the entire system.*

ZIP file	CSV file	Data saved in the file
	PHY_Cache_Usage_Rate_z.csv	Usage rates for cache memory in each MP unit.*
	PHY_Write_Pending_Rate_z.csv	Write pending rates in each MP unit.*
	PHY_Cache_Allocate_z.csv	The allocated size of the cache memory in each MP unit (MB).* This value does not correspond to the total capacity of cache because the value is the same as the allocated size of the cache memory that is managed by an MP unit.
• The letters “x-y” in CSV file names indicate a parity group or external volume group. • The letter “z” in CSV file names indicates the name of an MP unit. • (VSP 5000 series) Both long range and short range statistics are stored for resource usage and write pending rates. • (VSP 5000 series) You can select Long-Range or Short-Range from Data Range field in the Monitor Performance window. * This monitoring data indicates an instantaneous value at the time the data is collected.		

Parity group and external volume group statistics

The following table shows the file names and types of information in the Monitor Performance window that can be exported to files using Export Tool. These files contain statistics about parity groups and external volume groups.

ZIP file	CSV file	Data saved in the file
PG_dat.ZIP	PG_IOPS.csv	Number of read and write operations per second
	PG_TransRate.csv	Size of data transferred per second (KB/s)
	PG_Read_TransRate.csv	Size of read data transferred per second (KB/s)
	PG_Write_TransRate.csv	Size of write data transferred per second (KB/s)
	PG_Read_IOPS.csv	Number of read operations per second
	PG_Rnd_Read_IOPS.csv	Number of random read operations per second
	PG_CFW_Read_IOPS.csv (VSP 5000 series)	Number of read operations in "cache-fast-write" mode per second
	PG_Write_IOPS.csv	Number of write operations per second
	PG_Seq_Write_IOPS.csv	Number of sequential write operations per second
	PG_Rnd_Write_IOPS.csv	Number of random write operations per second
	PG_CFW_Write_IOPS.csv (VSP 5000 series)	Number of write operations in "cache-fast-write" mode per second
	PG_Read_Hit.csv	Read hit ratio
	PG_Seq_Read_Hit.csv	Read hit ratio in sequential access mode
	PG_Rnd_Read_Hit.csv	Read hit ratio in random access mode
	PG_CFW_Read_Hit.csv (VSP 5000 series)	Read hit ratio in "cache-fast-write" mode
	PG_Write_Hit.csv	Write hit ratio
	PG_Seq_Write_Hit.csv	Write hit ratio in sequential access mode

ZIP file	CSV file	Data saved in the file
	PG_Rnd_Write_Hit.csv	Write hit ratio in random access mode
	PG_CFW_Write_Hit.csv (VSP 5000 series)	Write hit ratio in "cache-fast-write" mode
	PG_BackTrans.csv	Number of data transfer operations between cache memory and data drives (for example, parity groups or external volume groups) per second
	PG_C2D_Trans.csv	Number of data transfer operations from cache memory and data drives (for example, parity groups or external volume groups) per second
	PG_D2CS_Trans.csv	Number of data transfer operations from data drives (for example, parity groups or external volume groups) per second to cache memory in sequential access mode
	PG_D2CR_Trans.csv	Number of data transfer operations from data drives (for example, parity groups or external volume groups) per second to cache memory in random access mode
	PG_Response.csv	Average response time (microsecond) at parity groups or external volume groups
	PG_Read_Response.csv	Average read response time (microsecond) at parity groups or external volume groups
	PG_Write_Response.csv	Average write response time (microsecond) at parity groups or external volume groups

Note: The parity group number is output in the column header of each performance value in these files.

Statistics for volumes in parity/external volume groups

The following table shows the file names and types of information in the Monitor Performance window that can be exported to files using Export Tool. These files contain statistics about volumes in parity groups or in external volume groups.

The letters "x-y" in CSV file names indicate a parity group. For example, if the file name is LDEV_IOPS_1-2.csv, the file contains the I/O rate for each volume in the parity group 1-2.

ZIP file	CSV file	Data saved in the file
LDEV_dat/LDEV_IOPS.ZIP	LDEV_IOPS_x-y.csv	Number of read and write operations per second
LDEV_dat/LDEV_TransRate.ZIP	LDEV_TransRate_x-y.csv	Size of data transferred per second (KB/s)
LDEV_dat/LDEV_Read_TransRate.ZIP	LDEV_Read_TransRate_x-y.csv	Size of read data transferred per second (KB/s)
LDEV_dat/LDEV_Write_TransRate.ZIP	LDEV_Write_TransRate_x-y.csv	Size of write data transferred per second (KB/s)
LDEV_dat/LDEV_Read_IOPS.ZIP	LDEV_Read_IOPS_x-y.csv	Number of read operations per second
LDEV_dat/LDEV_Seq_Read_IOPS.ZIP	LDEV_Seq_Read_IOPS_x-y.csv	Number of sequential read operations per second
LDEV_dat/LDEV_Rnd_Read_IOPS.ZIP	LDEV_Rnd_Read_IOPS_x-y.csv	Number of random read operations per second
LDEV_dat/LDEV_CFW_Read_IOPS.ZIP (VSP 5000 series)	LDEV_CFW_Read_IOPS_x-y.csv	Number of read operations in "cache-fast-write" mode per second
LDEV_dat/LDEV_Write_IOPS.ZIP	LDEV_Write_IOPS_x-y.csv	Number of write operations per second
LDEV_dat/LDEV_Seq_Write_IOPS.ZIP	LDEV_Seq_Write_IOPS_x-y.csv	Number of sequential write operations per second
LDEV_dat/LDEV_Rnd_Write_IOPS.ZIP	LDEV_Rnd_Write_IOPS_x-y.csv	Number of random write operations per second
LDEV_dat/LDEV_CFW_Write_IOPS.ZIP (VSP 5000 series)	LDEV_CFW_Write_IOPS_x-y.csv	Number of write operations in "cache-fast-write" mode per second
LDEV_dat/LDEV_Read_Hit.ZIP	LDEV_Read_Hit_x-y.csv	Read hit ratio

ZIP file	CSV file	Data saved in the file
LDEV_dat/LDEV_Seq_Read_Hit.ZIP	LDEV_Seq_Read_Hit_x-y.csv	Read hit ratio in sequential access mode
LDEV_dat/LDEV_Rnd_Read_Hit.ZIP	LDEV_Rnd_Read_Hit_x-y.csv	Read hit ratio in random access mode
LDEV_dat/LDEV_CFW_Read_Hit. ZIP (VSP 5000 series)	LDEV_CFW_Read_Hit_x-y.csv	Read hit ratio in "cache-fast-write" mode
LDEV_dat/LDEV_Write_Hit.ZIP	LDEV_Write_Hit_x-y.csv	Write hit ratio
LDEV_dat/LDEV_Seq_Write_Hit.ZIP	LDEV_Seq_Write_Hit_x-y.csv	Write hit ratio in sequential access mode
LDEV_dat/LDEV_Rnd_Write_Hit.ZIP	LDEV_Rnd_Write_Hit_x-y.csv	Write hit ratio in random access mode
LDEV_dat/LDEV_CFW_Write_Hit.ZIP (VSP 5000 series)	LDEV_CFW_Write_Hit_x-y.csv	Write hit ratio in "cache-fast-write" mode
LDEV_dat/LDEV_BackTrans.ZIP	LDEV_BackTrans_x-y.csv	Number of data transfer operations between cache memory and data drives (for example, volumes) per second
LDEV_dat/LDEV_C2D_Trans.ZIP	LDEV_C2D_Trans_x-y.csv	Number of data transfer operations from cache memory and data drives (for example, volumes) per second
LDEV_dat/LDEV_D2CS_Trans.ZIP	LDEV_D2CS_Trans_x-y.csv	Number of data transfer operations from data drives (for example, volumes) per second to cache memory in sequential access mode
LDEV_dat/LDEV_D2CR_Trans.ZIP	LDEV_D2CR_Trans_x-y.csv	Number of data transfer operations from data drives (for example, volumes) per second to cache memory in random access mode

ZIP file	CSV file	Data saved in the file
LDEV_dat/LDEV_Response.ZIP	LDEV_Response_x-y.csv	Average response time (microseconds) at volumes
LDEV_dat/LDEV_Read_Response.ZIP	LDEV_Read_Response_x-y.csv	Average read response time (microseconds) at volumes
LDEV_dat/LDEV_Write_Response.ZIP	LDEV_Write_Response_x-y.csv	Average write response time (microseconds) at volumes

Volumes in parity groups or external volume groups (at volumes controlled by a particular CU)

The following table shows the file names and types of information in the Monitor Performance window that can be exported to files using Export Tool. These files contain statistics about volumes in parity groups or external volume groups (at volumes controlled by a particular CU).

The letters "*xx*" in CSV file names indicate a CU number. For example, if the file name is LDEV_IOPS _10.csv, the file contains the I/O rate (per second) of the volumes controlled by the CU whose image number is 10.

ZIP file	CSV file	Data saved in the file
LDEVEachOfCU_dat/LDEV_Read_TransRate.ZIP	LDEV_Read_TransRate*xx*.csv	The size of read data transferred per second (KB/s)
LDEVEachOfCU_dat/LDEV_Write_TransRate.ZIP	LDEV_Write_TransRate*xx*.csv	The size of write data transferred per second (KB/s)
LDEVEachOfCU_dat/LDEV_Read_Response.ZIP	LDEV_Read_Response*xx*.csv	The average read response time (microseconds) at volumes
LDEVEachOfCU_dat/LDEV_Write_Response.ZIP	LDEV_Write_Response*xx*.csv	The average write response time (microseconds) at volumes
LDEVEachOfCU_dat/LDEV_IOPS.ZIP	LDEV_IOPS*xx*.csv	The number of read and write operations

ZIP file	CSV file	Data saved in the file
		per second
LDEVEachOfCU_dat/LDEV_TransRate.ZIP	LDEV_TransRate*xx*.csv	The size of data transferred per second (KB/s)
LDEVEachOfCU_dat/LDEV_Read_IOPS.ZIP	LDEV_Read_IOPS*xx*.csv	The number of read operations per second
LDEVEachOfCU_dat/LDEV_Seq_Read_IOPS.ZIP	LDEV_Seq_Read_IOPS*xx*.csv	The number of sequential read operations per second
LDEVEachOfCU_dat/LDEV_Rnd_Read_IOPS.ZIP	LDEV_Rnd_Read_IOPS*xx*.csv	The number of random read operations per second
LDEVEachOfCU_dat/LDEV_CFW_Read_IOPS.ZIP (VSP 5000 series)	LDEV_CFW_Read_IOPS*xx*.csv	The number of read operations in "cache-fast-write" mode per second
LDEVEachOfCU_dat/LDEV_Write_IOPS.ZIP	LDEV_Write_IOPS*xx*.csv	The number of write operations per second
LDEVEachOfCU_dat/LDEV_Seq_Write_IOPS.ZIP	LDEV_Seq_Write_IOPS*xx*.csv	The number of sequential write operations per second
LDEVEachOfCU_dat/LDEV_Rnd_Write_IOPS.ZIP	LDEV_Rnd_Write_IOPS*xx*.csv	The number of random write operations per second

ZIP file	CSV file	Data saved in the file
LDEVEachOfCU_dat/LDEV_CFW_Write_IOPS.ZIP (VSP 5000 series)	LDEV_CFW_Write_IOPS*xx*.csv	The number of write operations in "cache-fast-write" mode per second
LDEVEachOfCU_dat/LDEV_Read_Hit.ZIP	LDEV_Read_Hit*xx*.csv	The read hit ratio
LDEVEachOfCU_dat/LDEV_Seq_Read_Hit.ZIP	LDEV_Seq_Read_Hit*xx*.csv	The read hit ratio in sequential access mode
LDEVEachOfCU_dat/LDEV_Rnd_Read_Hit.ZIP	LDEV_Rnd_Read_Hit*xx*.csv	The read hit ratio in random access mode
LDEVEachOfCU_dat/LDEV_CFW_Read_Hit.ZIP (VSP 5000 series)	LDEV_CFW_Read_Hit*xx*.csv	The read hit ratio in "cache-fast-write" mode
LDEVEachOfCU_dat/LDEV_Write_Hit.ZIP	LDEV_Write_Hit*xx*.csv	The write hit ratio
LDEVEachOfCU_dat/LDEV_Seq_Write_Hit.ZIP	LDEV_Seq_Write_Hit*xx*.csv	The write hit ratio in sequential access mode
LDEVEachOfCU_dat/LDEV_Rnd_Write_Hit.ZIP	LDEV_Rnd_Write_Hit*xx*.csv	The write hit ratio in random access mode
LDEVEachOfCU_dat/LDEV_CFW_Write_Hit.ZIP (VSP 5000 series)	LDEV_CFW_Write_Hit*xx*.csv	The write hit ratio in "cache-fast-write" mode
LDEVEachOfCU_dat/LDEV_BackTrans.ZIP	LDEV_BackTrans*xx*.csv	The number of data transfer operations per second between cache memories and data drives (for example, volumes)

ZIP file	CSV file	Data saved in the file
LDEVEachOfCU_dat/LDEV_C2D_Trans.ZIP	LDEV_C2D_Trans*xx*.csv	The number of data transfer operations per second from cache memories and data drives (for example, volumes)
LDEVEachOfCU_dat/LDEV_D2CS_Trans.ZIP	LDEV_D2CS_Trans*xx*.csv	The number of data transfer operations per second from data drives (for example, volumes) to cache memories in sequential access mode
LDEVEachOfCU_dat/LDEV_D2CR_Trans.ZIP	LDEV_D2CR_Trans*xx*.csv	The number of data transfer operations per second from data drives (for example, volumes) to cache memories in random access mode
LDEVEachOfCU_dat/LDEV_Response.ZIP	LDEV_Response*xx*.csv	The average response time (microseconds) at volumes

Port statistics

The following table shows the file names and types of information in the Monitor Performance window that can be exported to files using Export Tool. These files contain statistics about ports.

Files with statistics about ports

ZIP file	CSV file	Data saved in the file
Port_dat.ZIP	Port_IOPS.csv	The number of read and write operations per second at Target ports
	Port_KBPS.csv	The size of data transferred per second at Target ports (KB/s)

ZIP file	CSV file	Data saved in the file
	Port_Response.csv	The average response time (microseconds) at Target ports
	Port_Initiator_IOPS.csv	The number of read and write operations per second at Initiator port
	Port_Initiator_KBPS.csv	The size of data transferred per second at Initiator port (KB/s)
	Port_Initiator_Response.csv	The average response time (microseconds) at Initiator port

Files with statistics about mainframe ports (VSP 5000 series)

ZIP file	CSV file	Data saved in the file
MF_Port_dat.ZIP	MF_Port_IOPS.csv	The number of read and write operations per second measured at every port
	MF_Port_Response.csv	Response time per port (in microseconds)
	MF_Port_Read_Write_KBPS.csv	Read/write data transfer amount per port (in KB/s)
	MF_Port_Read_KBPS.csv	Read data transfer amount per port (in KB/s)
	MF_Port_Write_KBPS.csv	Write data transfer amount per port (in KB/s)
	MF_Port_Avr_CMR.csv	Average CMR processing time (in microseconds)
	MF_Port_Avr_DisconnectTime.csv	Average disconnection time (in microseconds)
	MF_Port_Avr_ConnectTime.csv	Average connection time (in microseconds)
	MF_Port_Avr_OpenExchange.csv	Number of open exchanges per second

Host bus adapters connected to ports statistics

The following table shows the file names and types of information in the Monitor Performance window that can be exported to files using Export Tool. These files contain statistics about host bus adapters connected to ports.

The letters "*xx*" in CSV file names indicate a port name. For example, if the file name is PortWWN_1A_IOPS.csv, the file contains the I/O rate for each host bus adapter connected to the CL1-A port.

If files are exported to a Windows computer, CSV file names might end with numbers (for example, PortWWN_1A_IOPS-1.csv and PortWWN_1a_IOPS-2.csv).

ZIP file	CSV file	Data saved in the file
PortWWN_dat.ZIP	PortWWN_*xx*_IOPS.csv	The I/O rate (that is, the number of read and write operations per second) for HBAs that are connected to a port
	PortWWN_*xx*_KBPS.csv	The size of data transferred per second (KB/s) between a port and the HBAs connected to that port
	PortWWN_*xx*_Response.csv	The average response time (microseconds) between a port and the HBAs connected to that port

Volumes (LU) statistics

The following table shows the file names and types of information in the Monitor Performance window that can be exported to files using Export Tool. These files contain statistics about volumes (LUs).

ZIP file	CSV file	Data saved in the file
LU_dat.ZIP	LU_IOPS.csv	The number of read and write operations per second
	LU_TransRate.csv	The size of data transferred per second (KB/s)
	LU_Read_TransRate.csv	The size of read data transferred per second (KB/s)
	LU_Write_TransRate.csv	The size of write data transferred per second (KB/s)
	LU_Read_Response.csv	The average read response time (microseconds)
	LU_Write_Response.csv	The average write response time (microseconds)
	LU_Seq_Read_IOPS.csv	The number of sequential read operations per second
	LU_Rnd_Read_IOPS.csv	The number of random read operations per second

ZIP file	CSV file	Data saved in the file
	LU_Seq_Write_IOPS.csv	The number of sequential write operations per second
	LU_Rnd_Write_IOPS.csv	The number of random write operations per second
	LU_Seq_Read_Hit.csv	The read hit ratio in sequential access mode
	LU_Rnd_Read_Hit.csv	The read hit ratio in random access mode
	LU_Seq_Write_Hit.csv	The write hit ratio in sequential access mode
	LU_Rnd_Write_Hit.csv	The write hit ratio in random access mode
	LU_C2D_Trans.csv	The number of data transfer operations per second from cache memories to data drives (for example, LUs)
	LU_D2CS_Trans.csv	The number of data transfer operations per second from data drives (for example, LUs) to cache memories in sequential access mode
	LU_D2CR_Trans.csv	The number of data transfer operations per second from data drives (for example, LUs) to cache memories in random access mode
	LU_Response.csv	The average response time (microseconds) at volumes (LUs)

All host bus adapters connected to ports

The following table shows the file names and types of information in the Monitor Performance window that can be exported to files using Export Tool. These files contain statistics about all host bus adapters connected to ports.

The letters "*xx*" in CSV file names indicate the name of an SPM group. If files are exported to a Windows computer, CSV file names might end with numbers (for example, PPCGWWN_mygroup_IOPS-1.csv and PPCGWWN_MyGroup_IOPS-2.csv).

ZIP file	CSV file	Data saved in the file
PPCGWWN_dat.ZIP	PPCGWWN_*xx*_IOPS.csv	I/O rate (that is, the number of read and write operations per second) for HBAs belonging to an SPM group

ZIP file	CSV file	Data saved in the file
	PPCGWWN_*xx*_KBPS.csv	Transfer rate (KB/s) for HBAs belonging to an SPM group
	PPCGWWN_*xx*_Response.csv	Average response time (microseconds) for HBAs belonging to an SPM group
	PPCGWWN_NotGrouped_IOPS.csv	I/O rate (that is, the number of read and write operations per second) for HBAs that do not belong to any SPM group
	PPCGWWN_NotGrouped_KBPS.csv	Transfer rate (KB/s) for HBAs that do not belong to any SPM group
	PPCGWWN_NotGrouped_Response.csv	Average response time (microseconds), for HBAs that do not belong to any SPM group

MP units

(VSP 5000 series) The following table shows the file names and types of information in the Monitor Performance window that can be exported to files using Export Tool. The file contains statistics about usage rates of MPs.

The letter "*x*" in CSV file names indicates an MPPCB name and the letter "*y*" in CSV file names indicates an MP name.

ZIP file	CSV file	Data saved in the file
PhyProcDetail_dat.ZIP	PHY_MPU_x.y.csv	The MP usage rate of each resource allocated to MP units in short range is output in the following formats: • Performance information of LDEVs Kernel-type*;LDEV;LDEV-number;Usage-rate • Performance information of journals Kernel-type*;JNLG; Journal-number;Usage-rates • Performance information of external volumes Kernel-type*;ExG;External-volume-group-number;Usage-rate You can view up to 100 of the most used items in order of use. Caution: Use performance information as a guide to identify resources that greatly increase the MP usage rate. Adding the

ZIP file	CSV file	Data saved in the file
		performance items together does not equal the total estimated capacity of the MPs. Likewise, this performance information is not appropriate to estimate the usage of a particular resource.
* The kernel type is any one of the following types: Open-Target, Open-Initiator, Open-External, MF-Target, MF-External, BackEnd, or System.		

Remote copy operations by TC/TCz and monitoring data by GAD (whole volumes)

The following table shows the file names and types of information on the Usage Monitor tab in the TC, TCz, and GAD window that can be exported to files using Export Tool. These files contain statistics about remote copy operations (whole volumes) by TrueCopy, TrueCopy for Mainframe, and global-active device.

ZIP file	CSV file	Data saved in the file
RemoteCopy_dat.ZIP	RemoteCopy.csv	The following data in the whole volumes are saved: • The total number of remote I/Os (read and write operations) • The total number of remote write I/Os • The number of errors that occur during remote I/O • The number of initial copy remote I/Os • The average response time (milliseconds) for initial copy • The average transfer rate (KB/s) for initial copy remote I/Os • The number of update copy remote I/Os • The average transfer rate (KB/s) for update copy remote I/Os • The average response time (milliseconds) for update copy • The percentage of completion of copy operations (for example, number of synchronized pairs / total number of pairs)* • The number of tracks that have not yet been copied by the initial copy or resync copy operation*
* This monitoring data indicates an instantaneous value at the time the data is collected.		

Remote copy operations by TC and monitoring data by GAD (for each volume (LU))

The following table shows the file names and types of information on the Usage Monitor tab in the TC and GAD window that can be exported to files using Export Tool. These files contain statistics about remote copy operations (for each volume (LU)) by TrueCopy and global-active device. (VSP 5000 series) Note that this data cannot be obtained from a mainfame volume.

ZIP file	CSV file	Data saved in the file
RCLU_dat.ZIP	RCLU_All_RIO.csv	The total number of remote I/Os (read and write operations)
	RCLU_All_Write.csv	The total number of remote write I/Os
	RCLU_RIO_Error.csv	The number of errors that occur during remote I/O
	RCLU_Initial_Copy_RIO.csv	The number of initial copy remote I/Os
	RCLU_Initial_Copy_Transfer.csv	The average transfer rate (KB/s) for initial copy remote I/Os
	RCLU_Initial_Copy_Response.csv	The average response time (milliseconds) for the initial copy of each volume (LU)
	RCLU_Update_Copy_RIO.csv	The number of update copy remote I/Os
	RCLU_Update_Copy_Transfer.csv	The average transfer rate (KB/s) for update copy remote I/Os
	RCLU_Update_Copy_Response.csv	The average response time (milliseconds) for the update copy of each volume (LU)
	RCLU_Pair_Synchronized.csv	The percentage of completion of copy operations (for example, number of synchronized pairs / total number of pairs) *
	RCLU_Out_of_Tracks.csv	The number of tracks that have not yet been copied by the initial copy or resync copy operation *
* This monitoring data indicates an instantaneous value at the time the data is collected.		

Remote copy by TC/TCz and monitoring data by GAD (volumes controlled by a particular CU)

The following table shows the file names and types of information on the Usage Monitor tab in the TC, TCz, and GAD window that can be exported to files using Export Tool. These files contain statistics about remote copy operations (volumes controlled by a particular CU) by TrueCopy, TrueCopy for Mainframe, and global-active device.

The letters "*xx*" in CSV file names indicate a CU number. For example, if the file name is RCLDEV_All_RIO_10.csv, the file contains the total number of remote I/Os of the volumes controlled by the CU whose image number is 10.

ZIP file	CSV file	Data saved in the file
RCLDEV_dat/RCLDEV_All_RIO.ZIP	RCLDEV_All_RIO_*xx*.csv	The total number of remote I/Os (read and write operations)
RCLDEV_dat/RCLDEV_All_Write.ZIP	RCLDEV_All_Write_*xx*.csv	The total number of remote write I/Os
RCLDEV_dat/RCLDEV_RIO_Error.ZIP	RCLDEV_RIO_Error_*xx*.csv	The number of errors that occur during remote I/O
RCLDEV_dat/RCLDEV_Initial_Copy_RIO.ZIP	RCLDEV_Initial_Copy_RIO_*xx*.csv	The number of initial copy remote I/Os
RCLDEV_dat/RCLDEV_Initial_Copy_Transfer.ZIP	RCLDEV_Initial_Copy_Transfer_*xx*.csv	The average transfer rate (KB/s) for initial copy remote I/Os
RCLDEV_dat/RCLDEV_Initial_Copy_Response.ZIP	RCLDEV_Initial_Copy_Response_*xx*.csv	The average response time (milliseconds) for initial copy at volumes
RCLDEV_dat/RCLDEV_Update_Copy_RIO.ZIP	RCLDEV_Update_Copy_RIO_*xx*.csv	The number of update copy remote I/Os

ZIP file	CSV file	Data saved in the file
RCLDEV_dat/RCLDEV_Update_Copy_Transfer.ZIP	RCLDEV_Update_Copy_Transfer_*xx*.csv	The average transfer rate (KB/s) for update copy remote I/Os
RCLDEV_dat/RCLDEV_Update_Copy_Response.ZIP	RCLDEV_Update_Copy_Response_*xx*.csv	The average response time (milliseconds) for the update copy at volumes
RCLDEV_dat/RCLDEV_Pair_Synchronized.ZIP	RCLDEV_Pair_Synchronized_*xx*.csv	The percentage of completion of copy operations (for example, number of synchronized pairs / total number of pairs) *
RCLDEV_dat/RCLDEV_Out_of_Tracks.ZIP	RCLDEV_Out_of_Tracks_*xx*.csv	The number of tracks that have not yet been copied by the initial copy or Resync copy operation *
* This monitoring data indicates an instantaneous value at the time the data is collected.		

Remote copy by UR and URz (whole volumes)

The following table shows the file names and types of information on the Usage Monitor tab in the UR and URz window that can be exported to files using Export Tool. These files contain statistics about remote copy operations (whole volumes) by Universal Replicator and Universal Replicator for Mainframe.

ZIP file	CSV file	Data saved in the file
UniversalReplicator_dat.zip	UniversalReplicator.csv	The following data in the whole volumes are saved: • The number of write I/Os per second • The amount of data that are written per second (KB/s) • The initial copy hit rate (percent) • The average transfer rate (KB/s) for initial copy operations • The number of asynchronous remote I/Os per second at the primary storage system* • The number of journals at the primary storage system* • The average transfer rate (KB/s) for journals in the primary storage system* • The remote I/O average response time (milliseconds) on the primary storage system* • The number of asynchronous remote I/Os per second at the secondary storage system* • The number of journals at the secondary storage system* • The average transfer rate (KB/s) for journals in the secondary storage system* • The remote I/O average response time (milliseconds) on the secondary storage system*
* Includes the monitoring data at the time of executing initial copy.		

Remote copy by UR and URz (at journals)

The following table shows the file names and types of information on the Usage Monitor tab in the UR and URz window that can be exported to files using Export Tool. These files contain statistics about remote copy operations (at journals) by Universal Replicator and Universal Replicator for Mainframe.

ZIP file	CSV file	Data saved in the file
URJNL_dat.ZIP	URJNL_Write_Record.csv	The number of write I/Os from the host to the P-VOL per second
	URJNL_Write_Transfer.csv	The average transfer rate (KB/sec) of data transferred from the host to the primary storage system

ZIP file	CSV file	Data saved in the file
	URJNL_Initial_Copy_Hit.csv	The cache hit rate (percent) of the P-VOL during initial copy
	URJNL_Initial_Copy_Transfer.csv	The average transfer rate (KB/sec) for data transferred from the primary storage system during initial copy
	URJNL_M-JNL_Asynchronous_RIO.csv	The number of asynchronous remote I/Os[1] that the primary storage system receives from the secondary storage system per second
	URJNL_M-JNL_Asynchronous_Journal.csv	The total number[1] of journal data transferred from the primary storage system
	URJNL_M-JNL_Asynchronous_Copy_Transfer.csv	The average transfer rate[1] (KB/sec) of journal data transferred from ports used by UR and URz in the primary storage system
	URJNL_M-JNL_Asynchronous_Copy_Response.csv	The average time (average RIO response time)[1] (milliseconds) after the primary storage system receives asynchronous remote I/Os until the primary storage system responds to the secondary storage system
	URJNL_R-JNL_Asynchronous_RIO.csv	The total number[1] of asynchronous remote I/Os from the secondary storage system
	URJNL_R-JNL_Asynchronous_Journal.csv	The total number[1] of journal data received in the secondary storage system
	URJNL_R-JNL_Asynchronous_Copy_Transfer.csv	The average transfer rate[1] (KB/sec) for journal data received in the secondary storage system
	URJNL_R-JNL_Asynchronous_Copy_Response.csv	The average time (average RIO response time)[1] (milliseconds) after the secondary storage system sends asynchronous remote I/Os until the secondary storage

ZIP file	CSV file	Data saved in the file
		system receives responses from the primary storage system
	URJNL_M-JNL_Data_Used_Rate.csv	Data usage rate (percent) in an area where journal data of the primary storage system is stored[2]
	URJNL_M-JNL_Meta_Data_Used_Rate.csv	Metadata usage rate (percent) in an area where journal data of the primary storage system is stored[2]
	URJNL_R-JNL_Data_Used_Rate.csv	Data usage rate (percent) in an area where journal data of the secondary storage system is stored[2]
	URJNL_R-JNL_Meta_Data_Used_Rate.csv	Metadata usage rate (percent) in an area where journal data of the secondary storage system is stored[2]
Notes: 1. Monitoring data during initial copy is included. 2. This monitoring data indicates an instantaneous value at the time the data is collected.		

Remote copy by UR (for each volume (LU))

The following table shows the file names and types of information on the Usage Monitor tab in the UR window that can be exported to files using Export Tool. These files contain statistics about remote copy operations (for each volume (LU)) by Universal Replicator. (VSP 5000 series) Note that this data cannot be obtained from a mainfame volume.

ZIP file	CSV file	Data saved in the file
URLU_dat.ZIP	URLU_Read_Record.csv	The number of read I/Os from the host to the P-VOL per second
	URLU_Read_Hit.csv	The cash hit rate when the host reads data from the P-VOL
	URLU_Write_Record.csv	The number of write I/Os from the host to the P-VOL per second

ZIP file	CSV file	Data saved in the file
	URLU_Write_Hit.csv	The cash hit rate when the host writes data to the P-VOL
	URLU_Read_Transfer.csv	The average transfer rate (KB/sec) for data transferred when the host reads data from the P-VOL
	URLU_Write_Transfer.csv	The average transfer rate (KB/sec) for data transferred when the host writes data to the P-VOL
	URLU_Initial_Copy_Hit.csv	The cache hit rate (percent) of the P-VOL during initial copy
	URLU_Initial_Copy_Transfer.csv	The average transfer rate (KB/sec) for data transferred from the primary storage system during initial copy

Remote copy by UR and URz (at volumes controlled by a particular CU)

The following table shows the file names and types of information on the Usage Monitor tab in the UR and URz window that can be exported to files using Export Tool. These files contain statistics about remote copy operations (at volumes controlled by a particular CU) by Universal Replicator and Universal Replicator for Mainframe.

The letters "*xx*" in CSV file names indicate a CU number. For example, if the file name is URLDEV_Read_Record_10.csv, the file contains the number of read I/Os (per second) of the volumes controlled by the CU whose image number is 10.

ZIP file	CSV file	Data saved in the file
URLDEV_dat/URLDEV_Read_Record.ZIP	URLDEV_Read_Record_*xx*.csv	The number of read I/Os from the host to the P-VOL per second for a volume with CU number *xx*
URLDEV_dat/URLDEV_Read_Hit.ZIP	URLDEV_Read_Hit_*xx*.csv	The cache hit rate when the host reads data from the P-VOL for a volume

ZIP file	CSV file	Data saved in the file
		with CU number *xx*
URLDEV_dat/URLDEV_Write_Record.ZIP	URLDEV_Write_Record_*xx*.csv	The number of write I/Os from the host to the P-VOL per second for a volume with CU number *xx*
URLDEV_dat/URLDEV_Write_Hit.ZIP	URLDEV_Write_Hit_*xx*.csv	The cache hit rate when the host writes data to the P-VOL for a volume with CU number *xx*
URLDEV_dat/URLDEV_Read_Transfer.ZIP	URLDEV_Read_Transfer_*xx*.csv	The average transfer rate (KB/sec) for data transferred when the host reads data from the P-VOL for a volume with CU number *xx*
URLDEV_dat/URLDEV_Write_Transfer.ZIP	URLDEV_Write_Transfer_*xx*.csv	The average transfer rate (KB/sec) for data transferred when the host writes data to the P-VOL for a volume with CU number *xx*
URLDEV_dat/URLDEV_Initial_Copy_Hit.ZIP	URLDEV_Initial_Copy_Hit_*xx*.csv	The cache hit rate of the P-VOL for initial copy (percent), for each volume identified by CU number *xx*

ZIP file	CSV file	Data saved in the file
URLDEV_dat/URLDEV_Initial_Copy_Transfer.ZIP	URLDEV_Initial_Copy_Transfer_*xx*.csv	The average transfer rate (KB/sec) for data transferred from the primary storage system during initial copy for a volume with CU number *xx*

Causes of invalid monitoring data

If the value of monitoring data in CSV files is less than 0 (zero), consider the following causes:

Invalid values of monitoring data	Probable causes
The monitoring data in the CSV file includes (-1).	The value (-1) indicates that Performance Monitor failed to obtain monitoring data. Probable reasons are: • Performance Monitor attempted to obtain statistics when an operation for restarting the storage system is in progress. • Performance Monitor attempted to obtain statistics when a heavy workload is imposed on the storage system. • There is no volume in a parity group. • Just after the CUs to be monitored were added, Export Tool failed to save files that contain monitoring data for all volumes or journal volumes used by remote copy software (TC, TCz, UR, URz) and GAD. For details about the files, see Remote copy operations by TC/TCz and monitoring data by GAD (whole volumes), Remote copy by UR and URz (whole volumes), and Remote copy by UR and URz (at journals). • (VSP 5000 series) If no CU is specified as a monitoring target, the value (-1) is displayed as the monitoring data. • (VSP 5000 series) If Disable is selected to stop monitoring in the Monitoring Switch field on the Monitoring Options window and longrange is specified as the sampling interval, the monitoring data for the period when Performance Monitor stops monitoring is (-1). • (VSP 5000 series) If you added the CU during monitoring, specified longrange as the sampling interval, and collected monitoring data, the value (-1) is displayed as the monitoring data before the CU was added. • (VSP 5000 series) If the CU number is not the monitoring target object, Performance Monitor cannot obtain monitoring data from the CU. However, when the RemoteCopy, Universal Replicator, or URJNL operand is specified for the group subcommand, the value (-1) is not displayed as the monitoring data even if the CU number is not the monitoring target object.

Invalid values of monitoring data	Probable causes
	In that case, data on the monitored CU is added up and output to the CSV file. • (VSP 5000 series) If all CU numbers are not the monitoring target, the value (0) is displayed as the monitoring data. • (VSP 5000 series) If the mainframe host runs a command to clear usage information, the monitoring data counter is cleared. As a result, the monitoring data at that point becomes disabled, and the value (-1) displays as the monitoring data. • When time synchronization with the SNTP server or auto summer time is enabled on the SVP, the time is automatically adjusted. If the adjusted time difference is large, the invalid value (-1) might be output as the monitoring data because correct monitoring data cannot be obtained. • If correct monitoring data cannot be obtained because the storage system is undergoing a certain maintenance operation, the value (-1) is output as the monitoring data. • (VSP E series) Performance Monitor could not obtain monitoring data due to a real-time virus scan by a virus detection program installed on the SVP. Exclude the Device Manager - Storage Navigator installation directory from the real-time virus scan target. For more information, see the *System Administrator Guide*.
The monitoring data in the CSV file includes (-3).	The value (-3) indicates that Performance Monitor failed to obtain monitoring data for the following reason: If IOPS is 0 (zero), the Response Time that is included in the monitoring data for LUs, LDEVs, ports, WWNs, or external volumes is (-3). For mainframe ports, the Average CMR Processing Time, Average Disconnection Time, and Average Connection Time are also (-3). Because IOPS is 0 (zero), these values become invalid.
The monitoring data in the CSV file includes (-4).	The value (-4) indicates that Performance Monitor failed to obtain monitoring data for the following reason: If the period for the monitoring data that is specified with Export Tool does not match the collecting period for monitoring data, Export Tool cannot collect the monitoring data. If data of SVP is updated while the monitoring data is being collected, the collected monitoring data near the collection start time is (-4).
The monitoring data in the CSV file includes "-5".	When the CU number is not the monitoring target object, Performance Monitor cannot obtain monitoring data from the CU. If the PG, LDEV, LU, RCLU, RCLDEV, URLU, or URLDEV operand is specified, the value of the monitoring data is "-5". To solve this problem, specify the CU as the monitoring target object by using the Monitoring Options window of Performance Monitor (not by using Export Tool). If the RemoteCopy, Universal Replicator, or URJNL operand is specified, the value "-5" is not output in the monitoring data though the CU number is not the

Invalid values of monitoring data	Probable causes
	monitoring target object. In this case, data on monitored CUs are summed up and output into the CSV file.